TECHNISCHE UNIVERSITÄT DRESDEN

DRESDEN concept

MOD-SEM SuSe25

Topic 3 - Phase 2

Hyperuniformity

Generation of hyperuniform structures and their analysis

Contents - Phase 2

Recap: Hyperuniformity

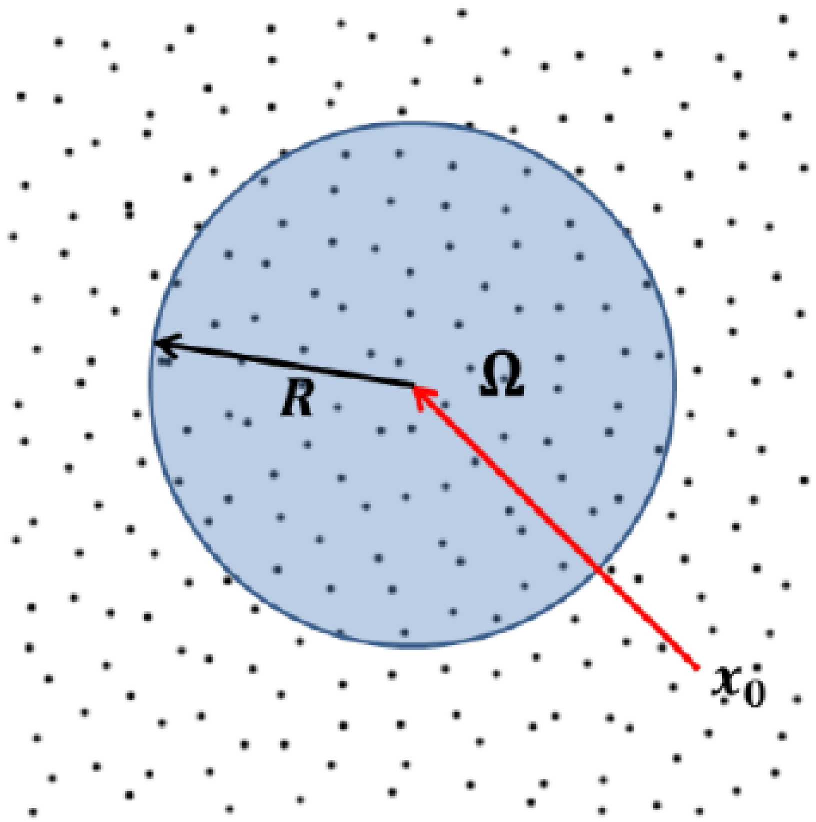
- **Hyperuniform systems are defined by the suppression of density fluctuations over large spatial scales.**

- One can derive from internal correlation and fourier transform [Tor18] that hyperuniformity may be characterized by a vanishing structure factor for small wave vectors

$$\lim_{|k| \to 0} S(k) = 0$$

- For a single periodic configuration with a finite number of N points, the structure factor is identical to [Tor18]:

$$S(k) = \frac{1}{N} |\mathcal{F}(\sum_{j=1}^{N} \delta(r - r_j))|^2 = \frac{1}{N} | \sum_{j=1}^{N} 1 \cdot \exp(ik \cdot r_j)|^2, k \in \mathcal{K}$$



*[MT17]

Generating hyperuniform structures

Generation of hyperuniform point patterns using the collective coordiante approach

Generating hyperuniform structures
Generation of hyperuniform point patterns

- We may use a method based on optimization [SSDV24] towards a given structure factor

$$\min_r \mathcal{L}(r) = \min_{r_1,\ldots,r_N} \sum_{k\in\mathcal{K}} |S(k) - S_0(k)|^2$$

- We can compute the derivative of the Loss analytically

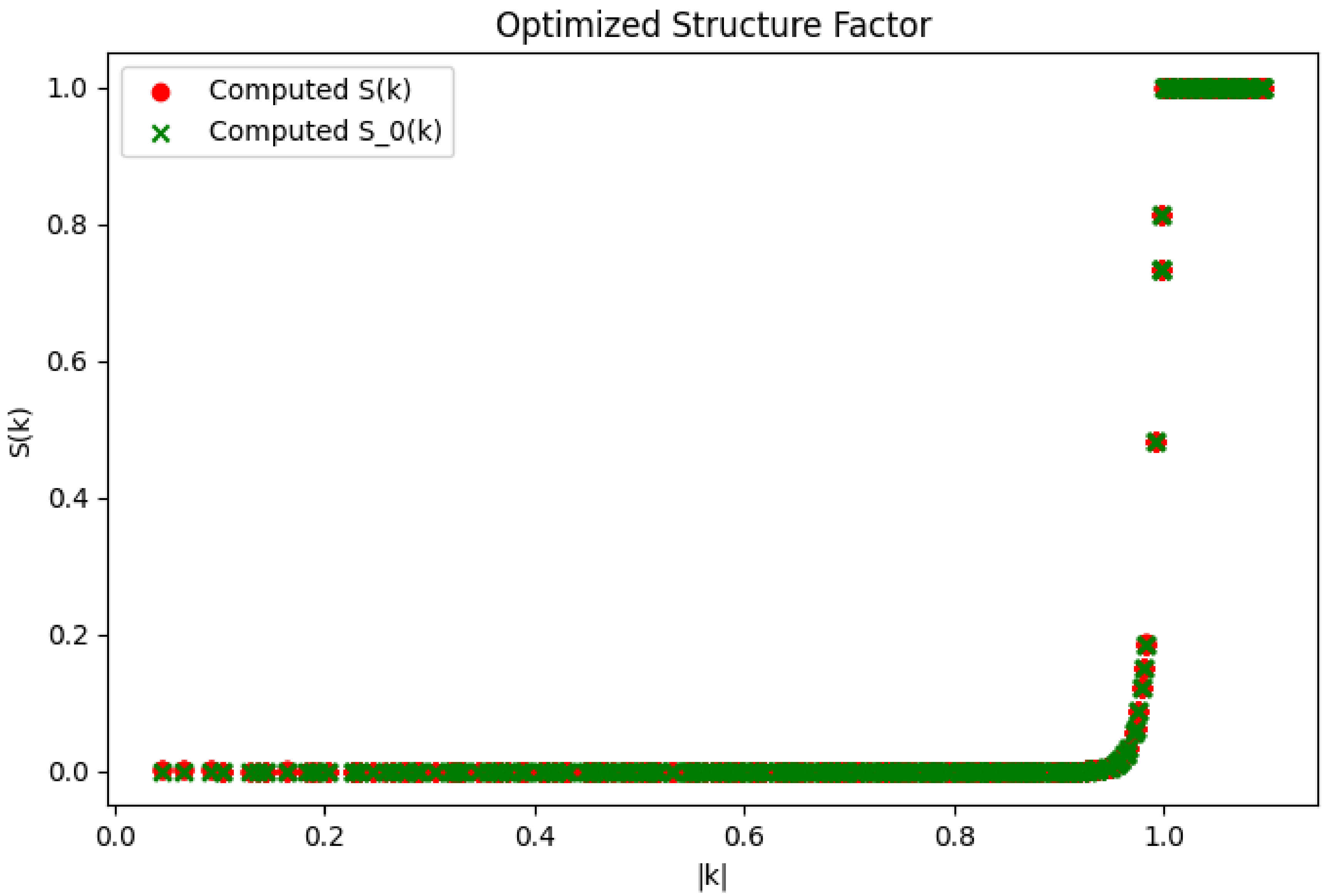
$$\frac{d\mathcal{L}}{dr_i} = \mathrm{Re}\Big(\sum_{k\in\mathcal{K}} \frac{-4\mathbf{i}kw(k)}{N}(S(k,\mathbf{r}) - S_0(k))\widehat{\rho}\exp(-\mathbf{i}k\cdot r_i)\Big)$$

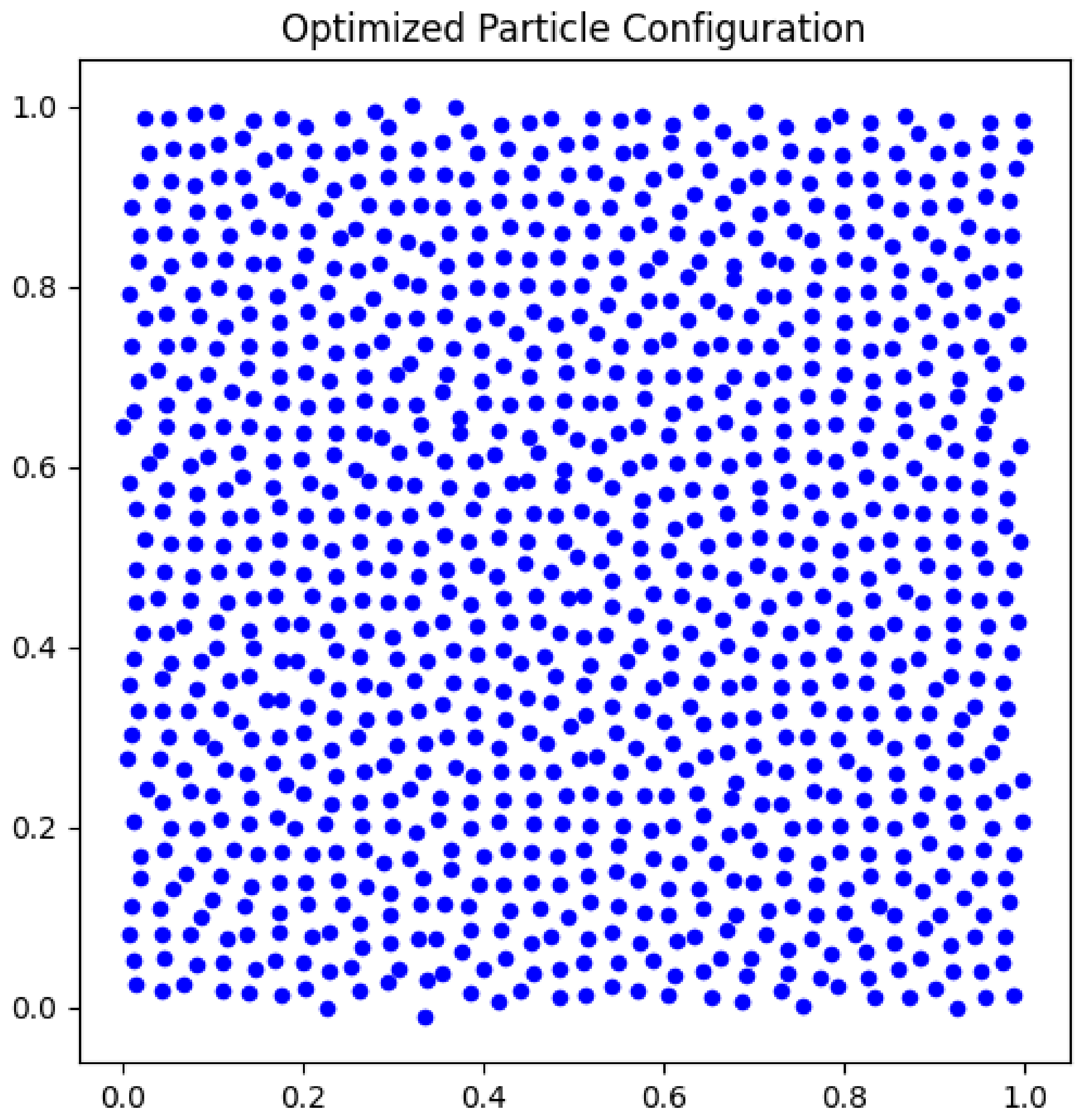
- And feed it into a minimizer of our choice (e.g. SciPy's BFGS) together with an initial point configuration (e.g. randomly perturbed grid)
- We may parametrize

$$S_0(k) = \begin{cases} K^{-\alpha}(1-H)|k|^{\alpha} + H, & |k| < K \\ 1, & \text{else} \end{cases}$$

Generating hyperuniform structures
Generation of hyperuniform point patterns

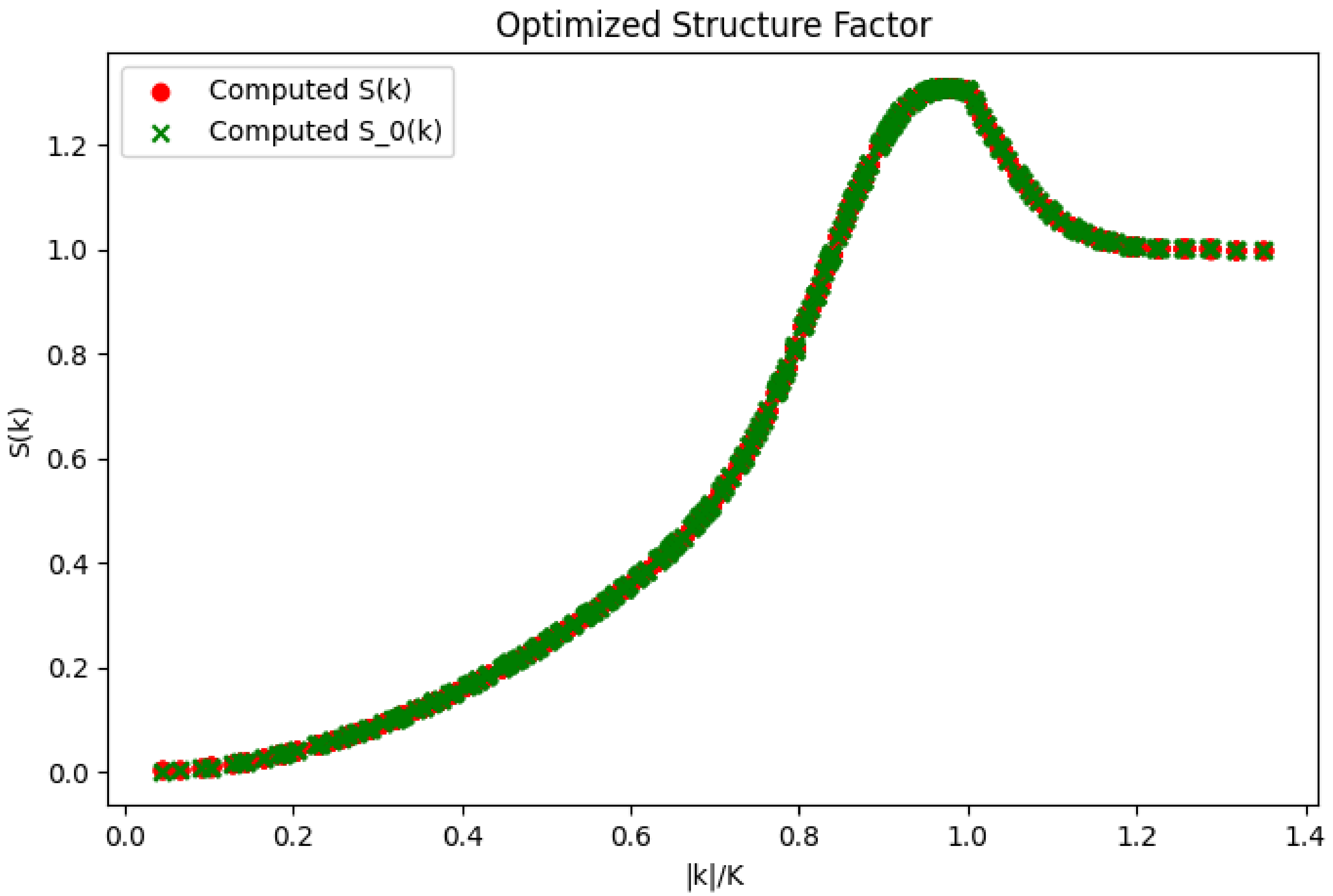
- We can easily generate different point patterns

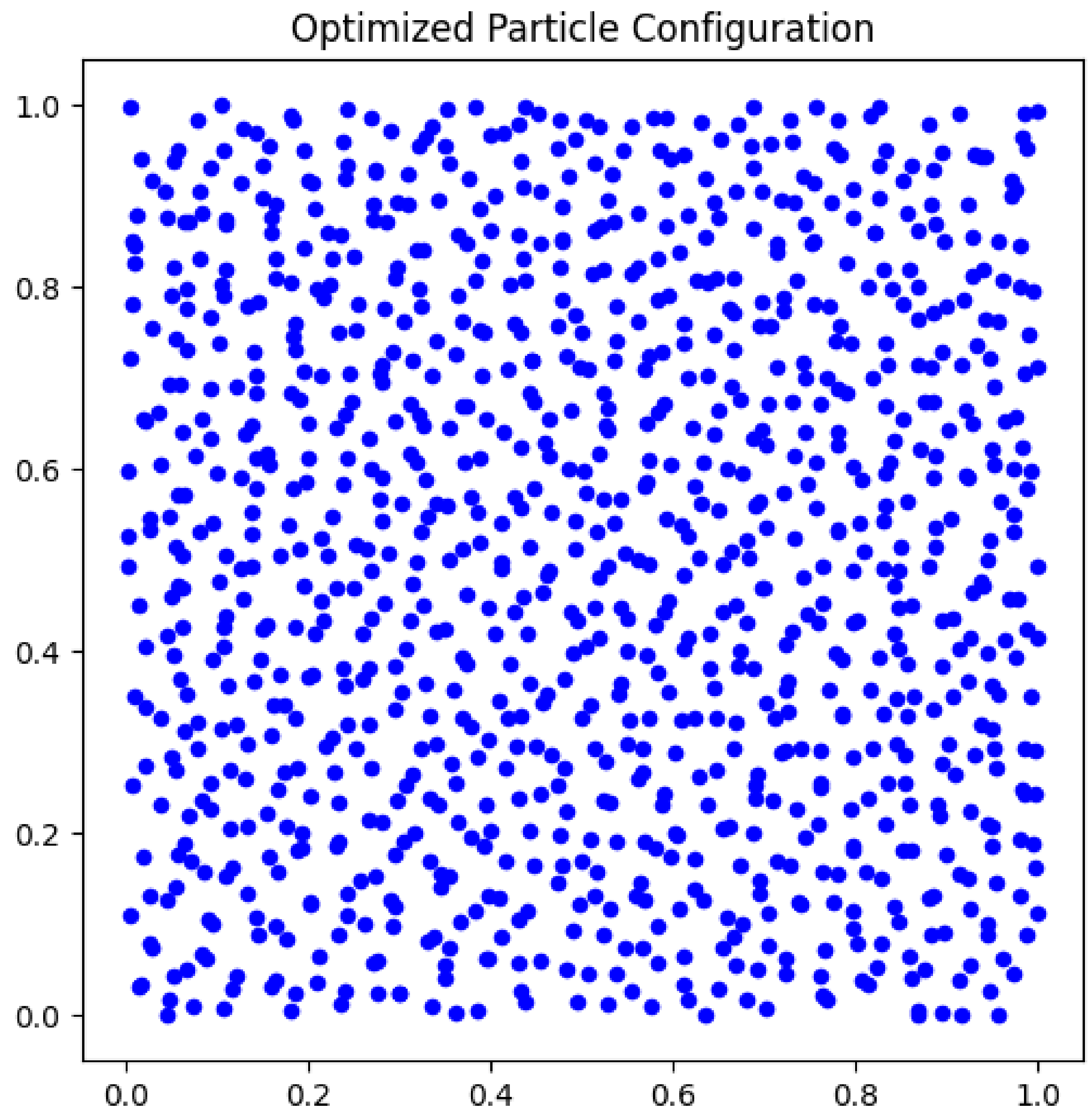




Generating hyperuniform structures
Generation of hyperuniform point patterns

- We can easily generate different point patterns





Generating hyperuniform structures

Acceleration using nuFFT

Generating hyperuniform structures

Generation of hyperuniform point patterns - Acceleration using nuFFT

- In [SCM24] a scalable nuFFT-based approach to generating HU point patterns is proposed
- nuFFT is used to evaluate expressions of the form

$$f_j = \sum_{i=1}^{I} c_i \exp(\pm \mathbf{i} y_j \cdot x_i), \; j = 1, \ldots, J$$

- where $x_i \in \mathbb{R}^d, c_i \in \mathbb{C}, y_j \in \mathbb{R}^d$ in $\mathcal{O}(N \log N)$ [FIN24]
- Recall:

$$S(k, r) = \frac{1}{N} | \sum_{i=1}^{N} 1 \cdot \exp(ik \cdot r_i) |^2$$

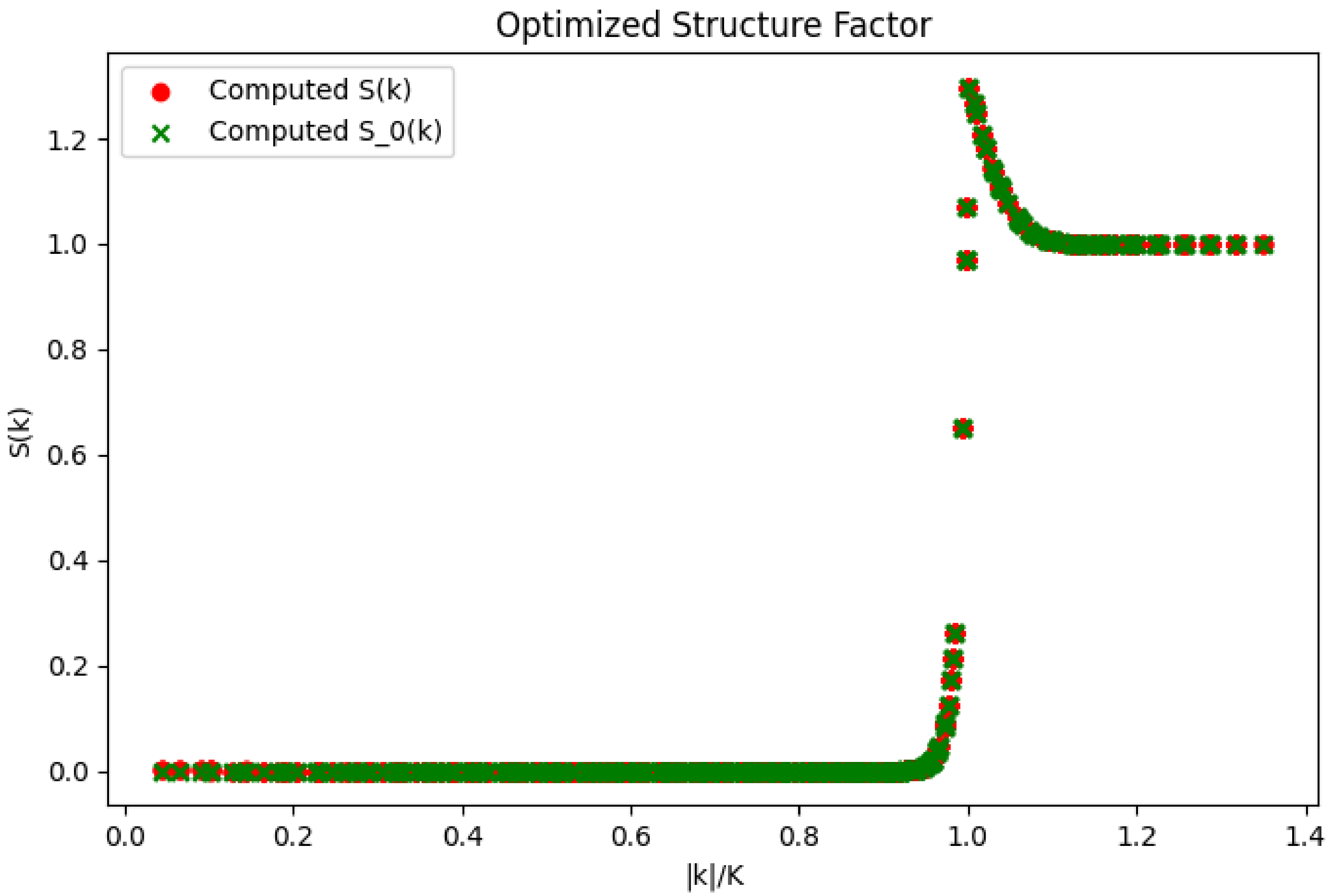
$$\frac{d\mathcal{L}}{dr_i} = \mathrm{Re}(\sum_{k \in \mathcal{K}} \frac{-4\mathbf{i}kw(k)}{N} (S(k, \mathbf{r}) - S_0(k)) \widehat{\rho} \exp(-\mathbf{i}k \cdot r_i))$$

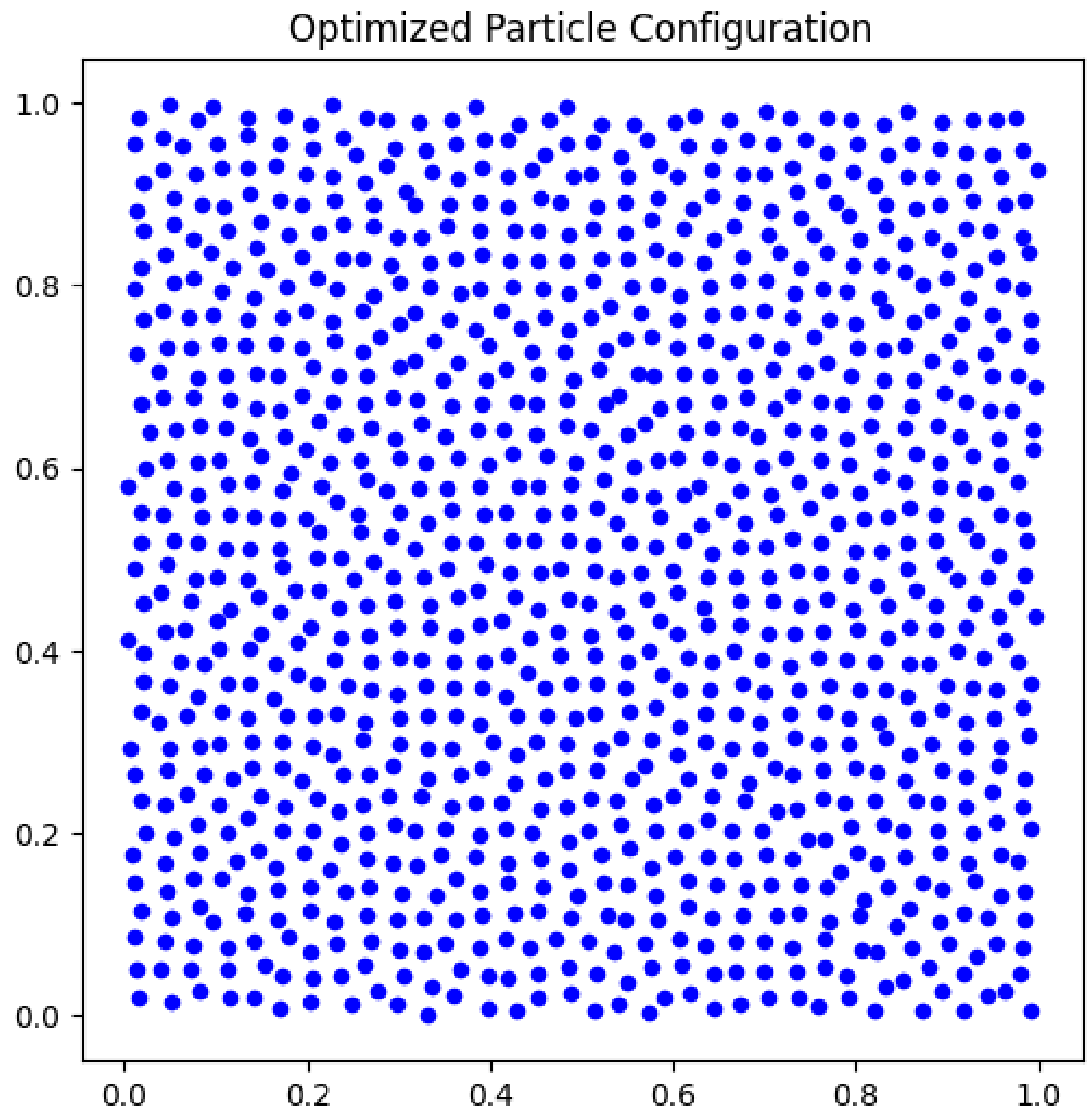
- Therefore we can compute the structure factor and the derivative of the loss function using nuFFT

Generating hyperuniform structures

Generation of hyperuniform point patterns - Acceleration using nuFFT

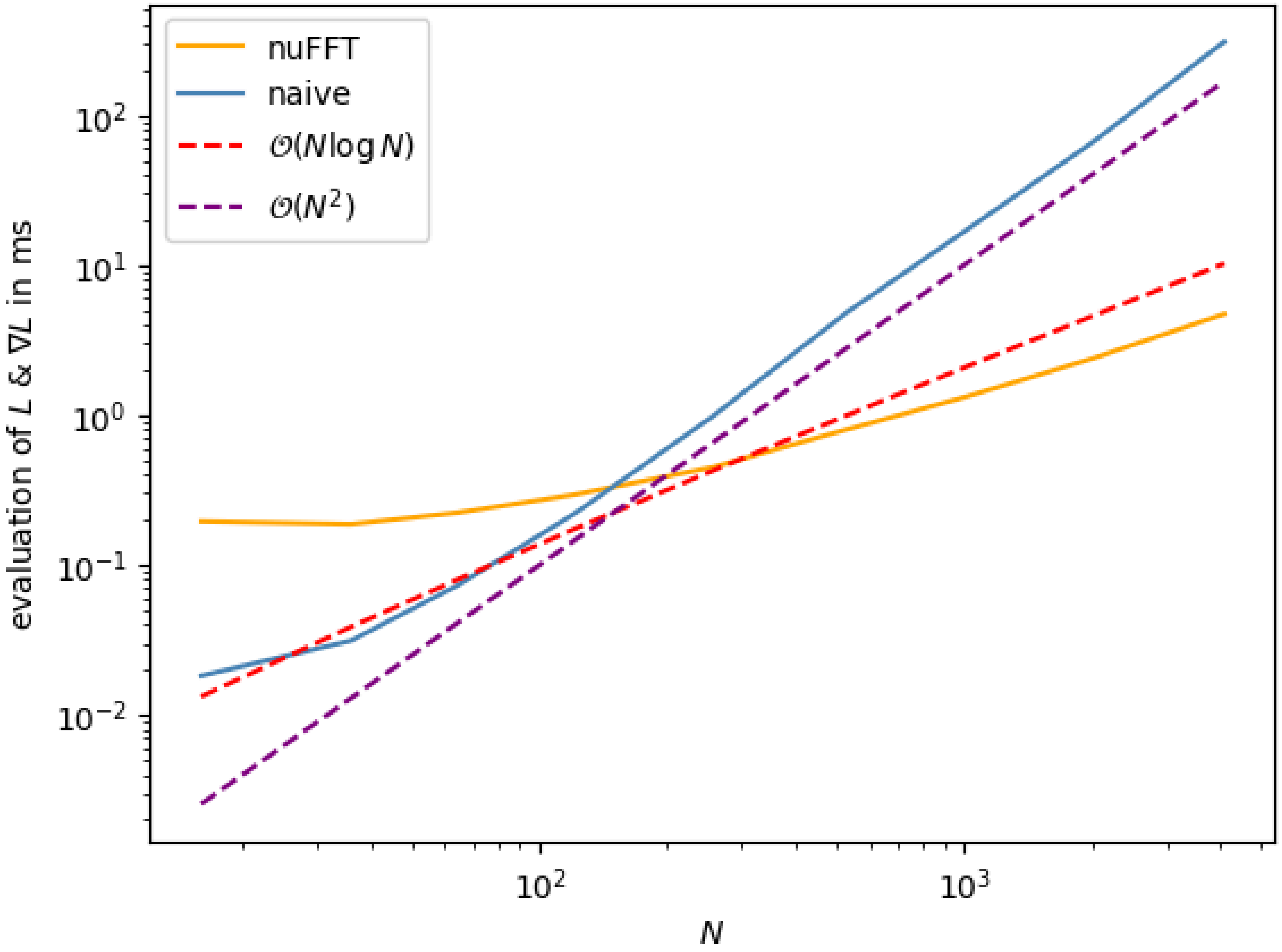
- We can once again generate point patterns





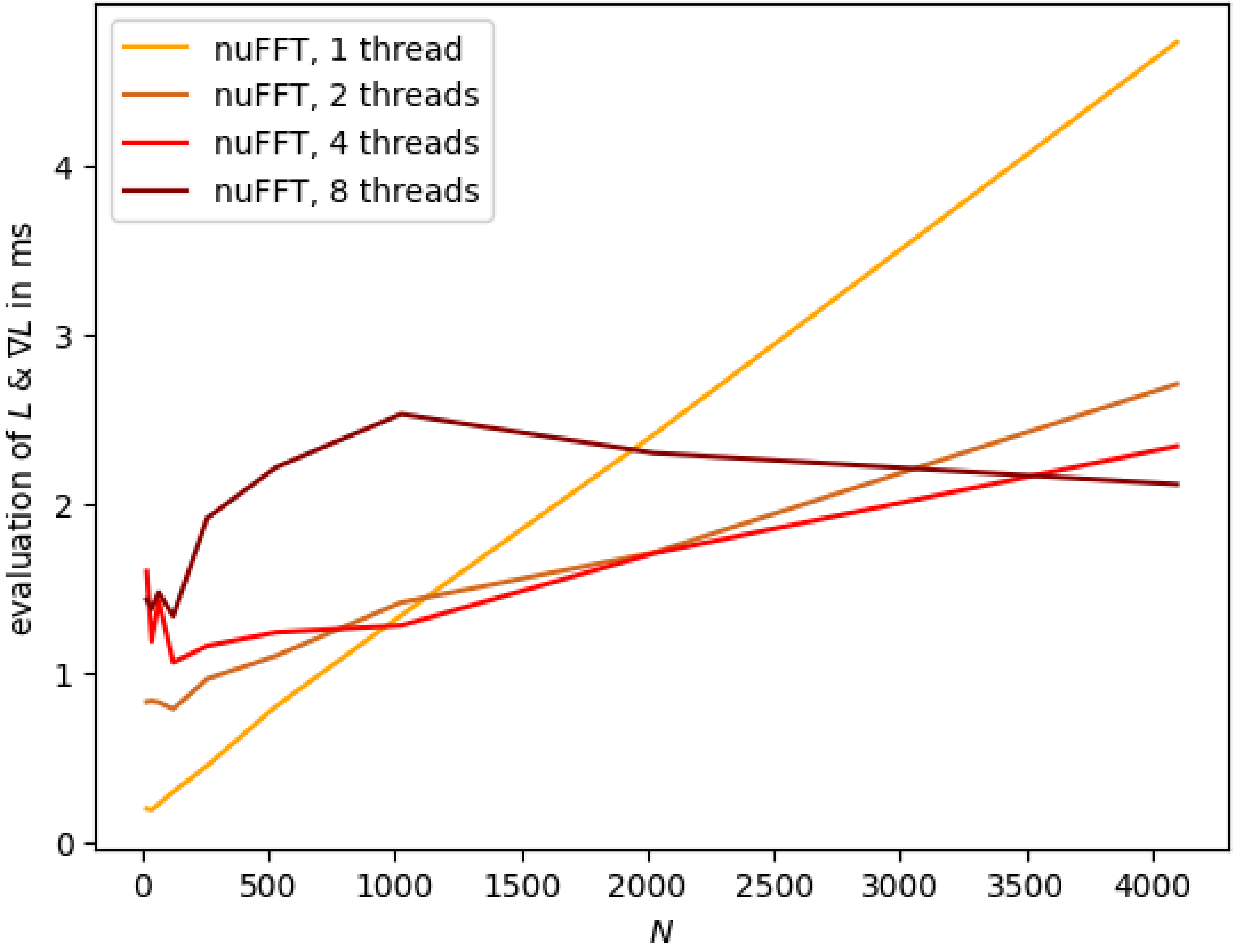
Generating hyperuniform structures
Acceleration using nuFFT - Benchmark

- We benchmarked the two methods



Generating hyperuniform structures
Benchmark - Multithreading

- We benchmarked the method with multithreading



Generating hyperuniform structures
Benchmark - Full Simulation

- We observe faster generation times of patterns of $N = 1024$ points ($M = 1652$ wave vectors)

	naive	nuFFT
Function evaluations	1519	1004
Gradient evaluations	1519	1004
Iterations	1000	1000
Loss value	$\approx 2.6 \cdot 10^{-4}$	$\approx 3.0 \cdot 10^{-4}$
Time	11m 24.2s	6m 47.6s

Generating hyperuniform structures

Generation of hyperuniform gaussian scalar fields

Generating hyperuniform structures
HU scalar fields

- Hyperuniformity can be extended to scalar fields
- We may be able to extend analysis of HU point patterns to analysis of HU scalar fields

Task: Generate hyperuniform scalar field from a gaussian random field [MT17], Section III & IV

Generating hyperuniform structures
HU scalar fields

- Gaussian random fields can be built by summing wave vector components that satisfy a specific autocovariance condition. To create hyperuniform scalar fields, one can choose amplitude functions **A(k)** that vanish at **k=0**, ensuring the spectral density also vanishes there.

$$\boxed{f(\mathbf{r}) = C \sum_{i=1}^{N} A(k_i) \cos(\mathbf{k}_i \cdot \mathbf{r} + \phi_i) \Delta k_i^d}$$

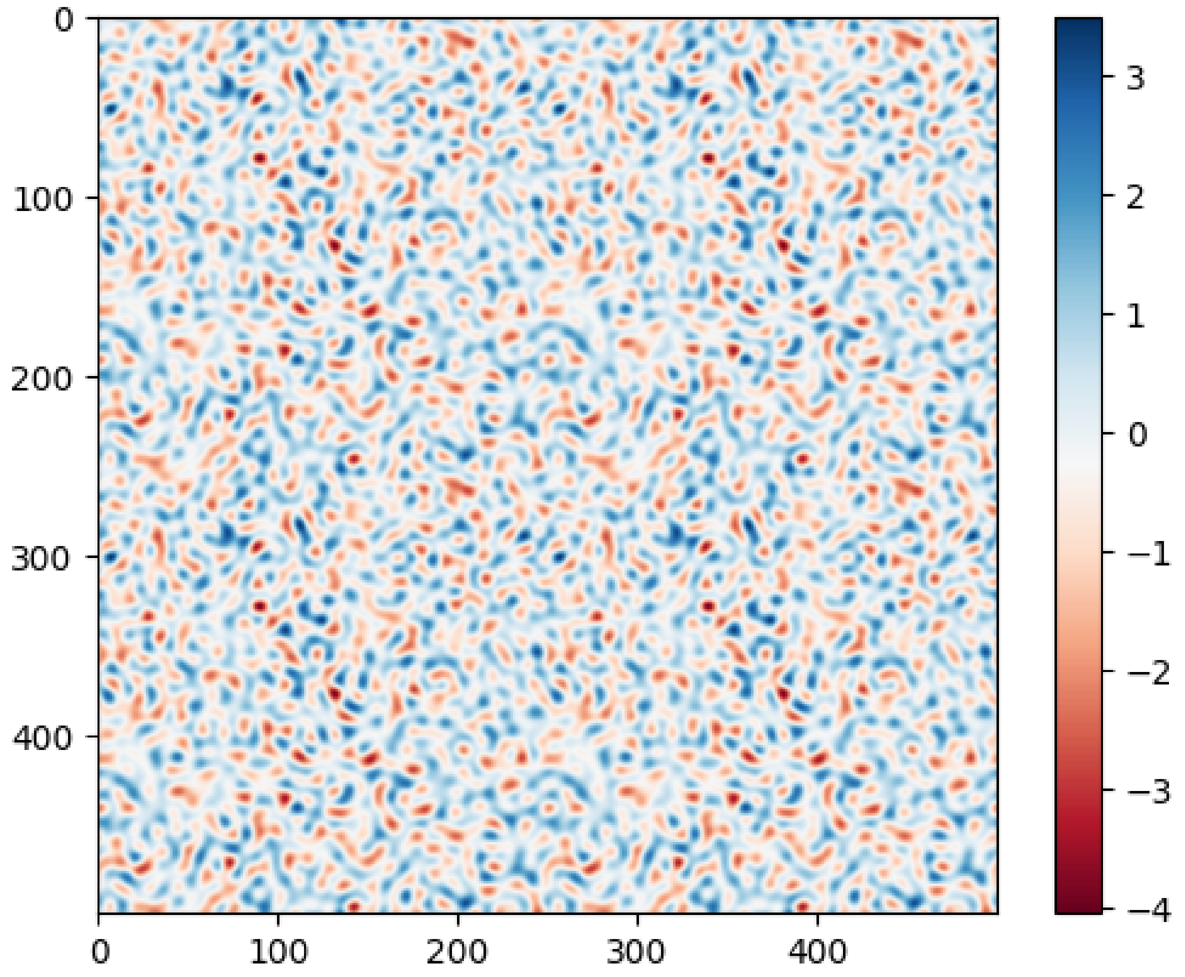
$$A(k) = \sqrt{k} e^{-\alpha k^2}$$

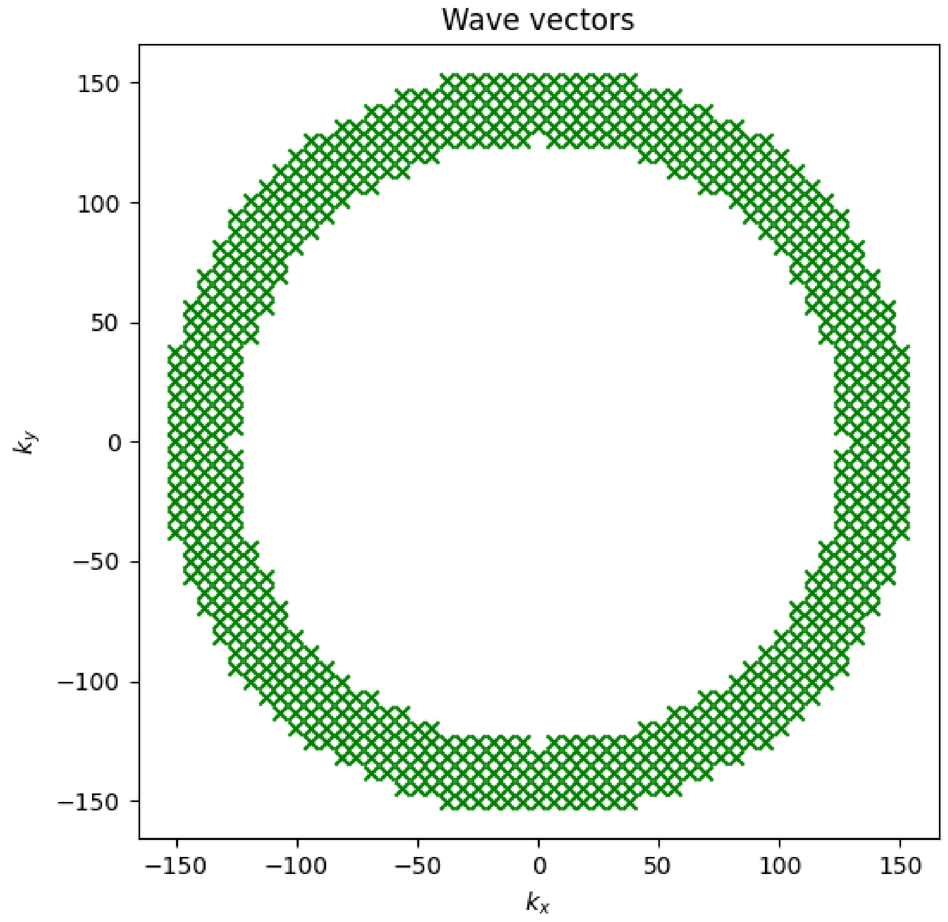
$$A(k) = \frac{k^2}{\sqrt{\beta + k^{10}}}$$

Generating hyperuniform structures
HU scalar fields

$$\boxed{f(\mathbf{r}) = C\sum_{i=1}^{N} A(k_i)\cos(\mathbf{k}_i\cdot\mathbf{r}+\phi_i)\Delta k_i^d}$$

$$A(k) = \sqrt{k}e^{-\alpha k^2}$$





Generating hyperuniform structures
HU scalar fields

$$\boxed{f(\mathbf{r}) = C \sum_{i=1}^{N} A(k_i) \cos(\mathbf{k}_i \cdot \mathbf{r} + \phi_i) \Delta k_i^d}$$

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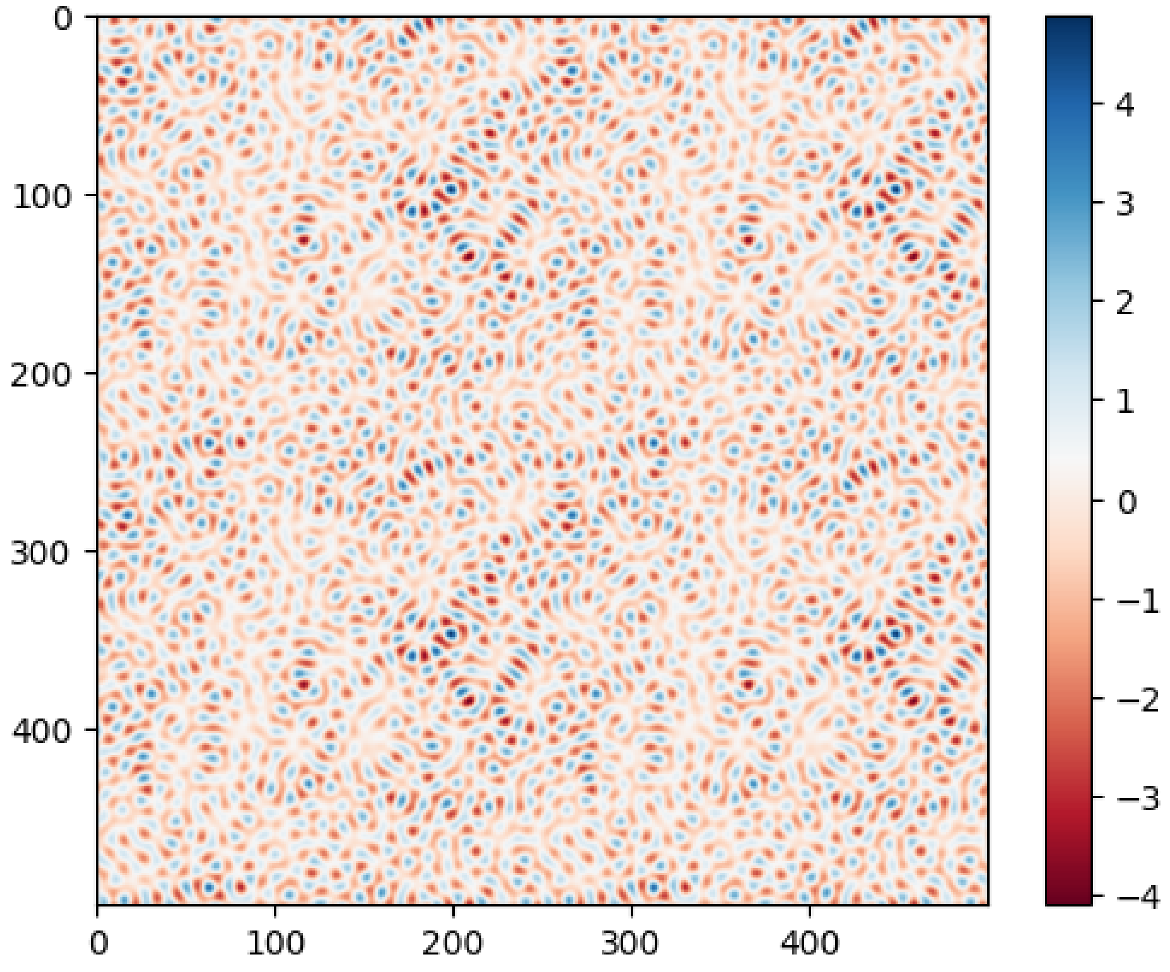
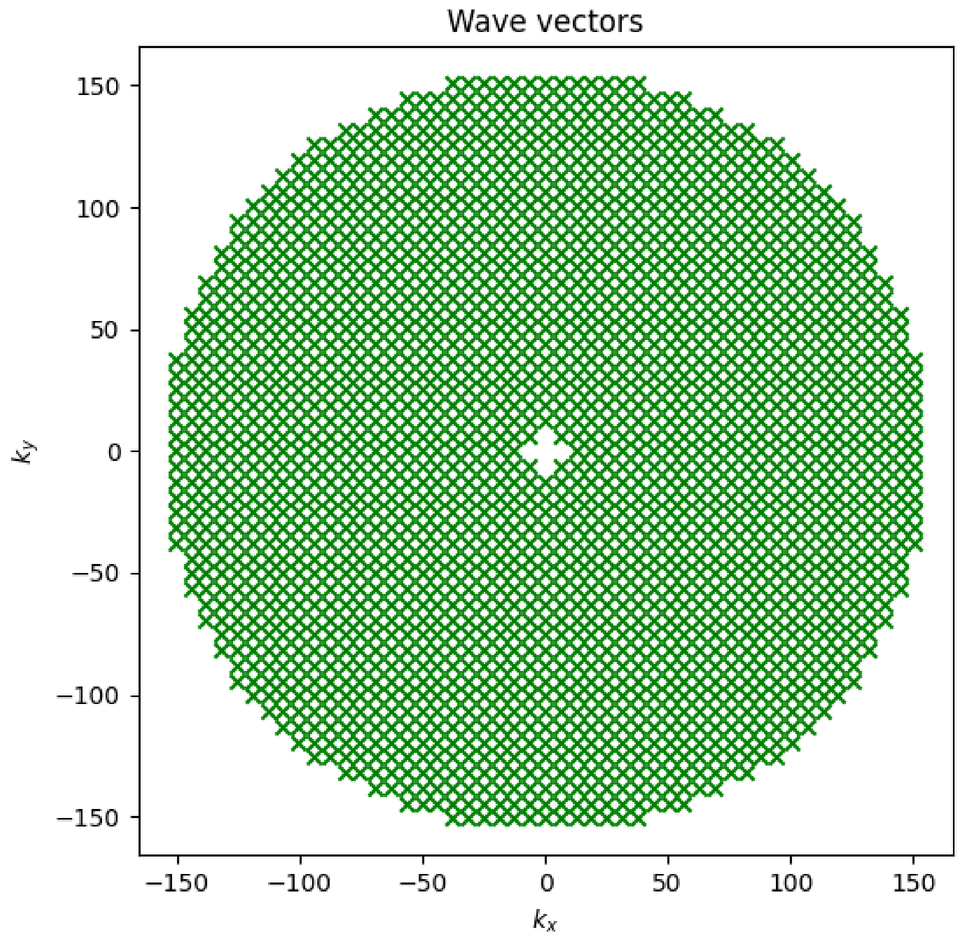
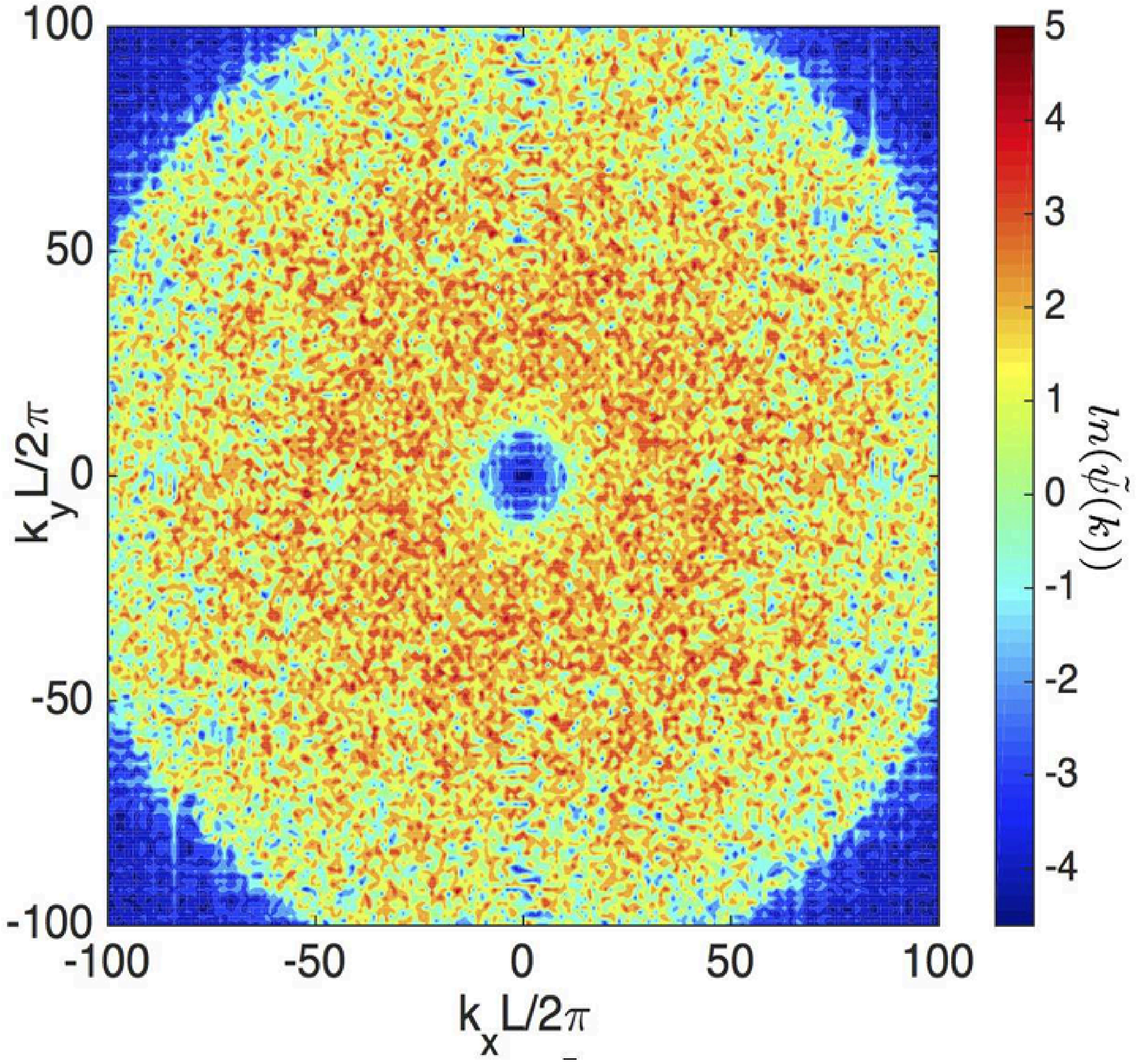


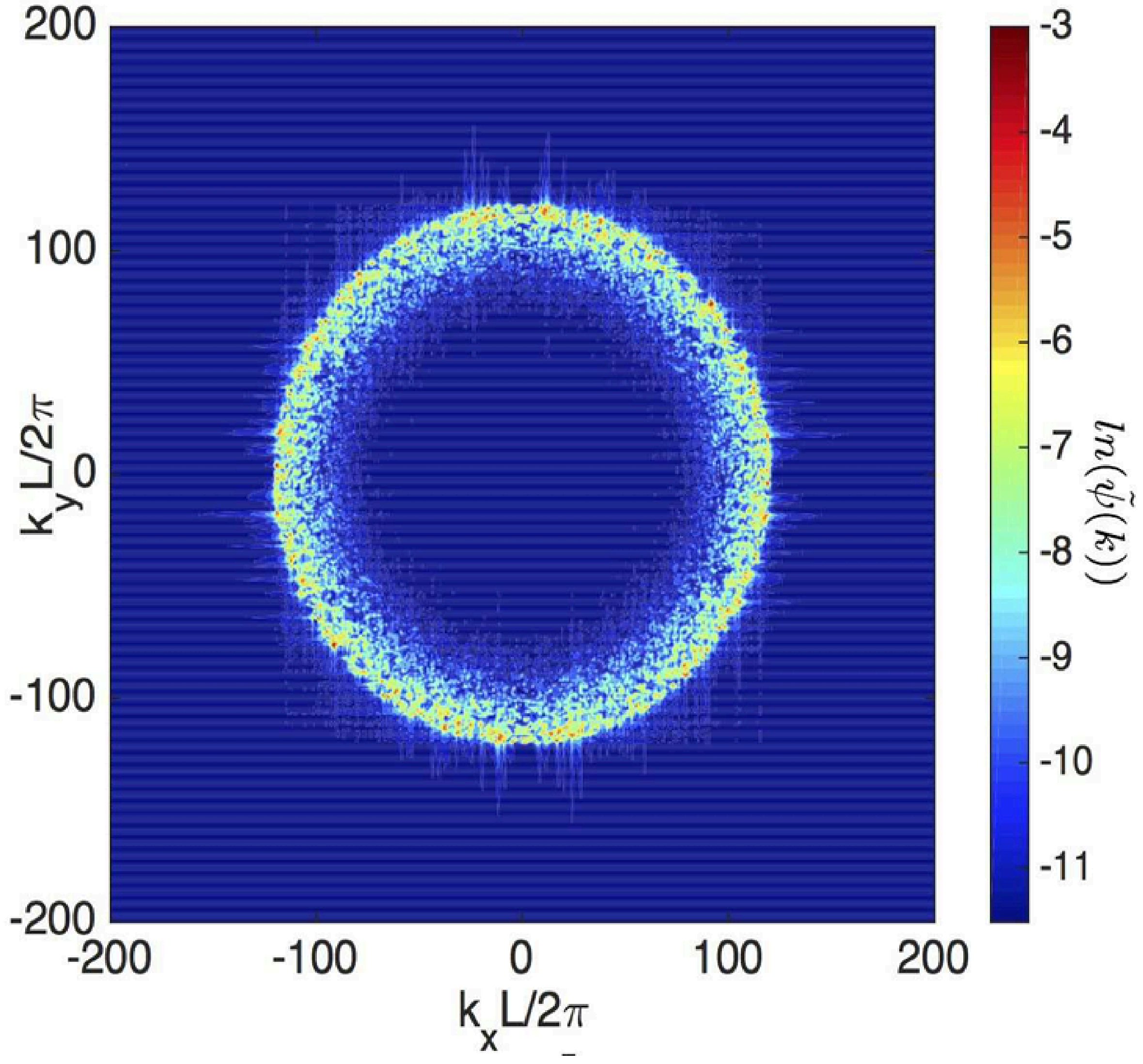


Generating hyperuniform structures
HU scalar fields

$$A(k) = \sqrt{k}e^{-\alpha k^2}$$

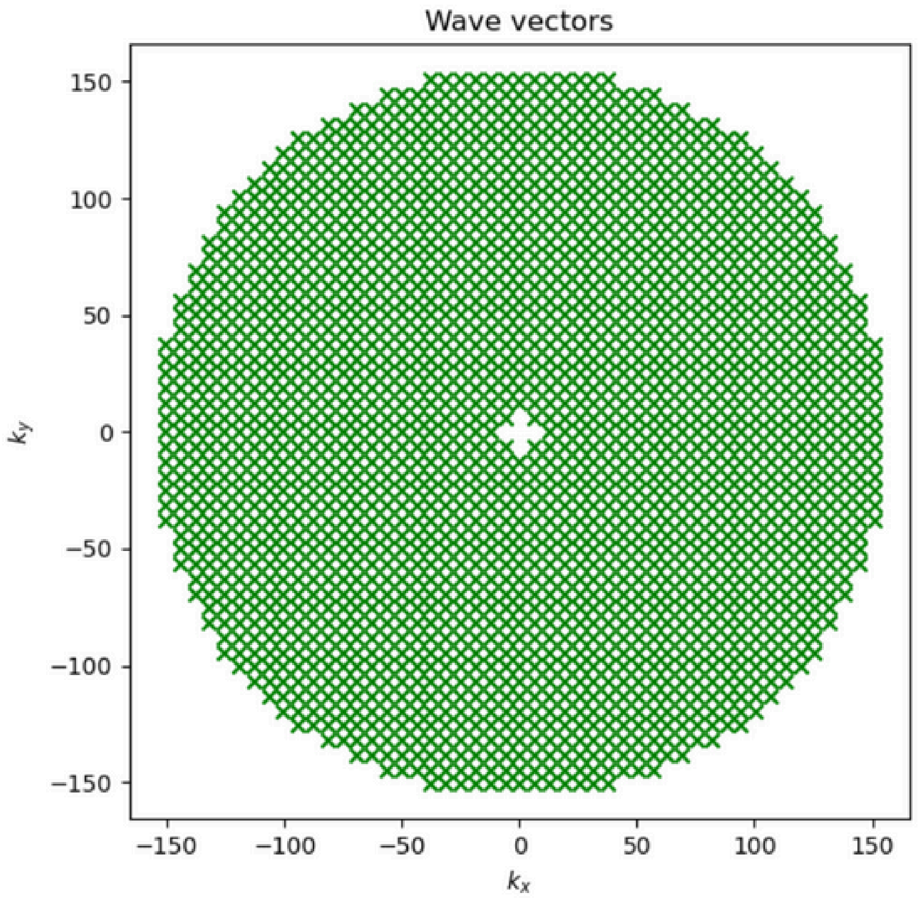


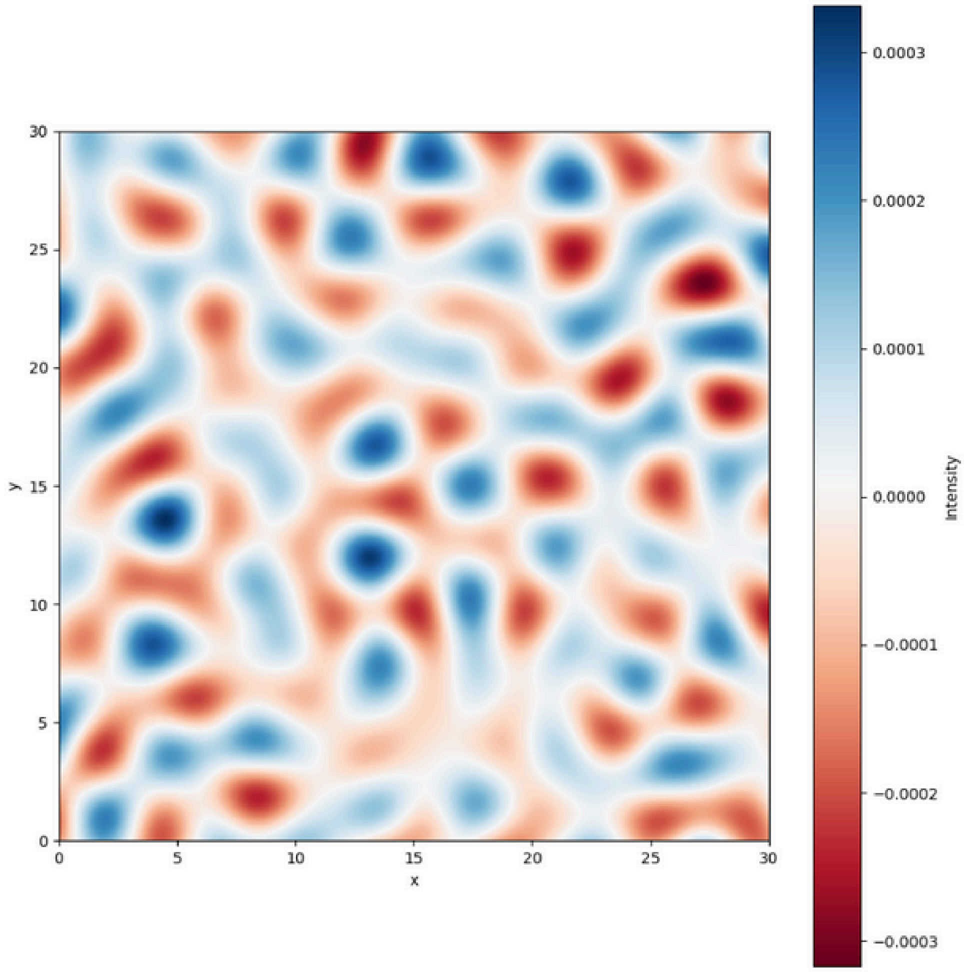
$$A(k) = \frac{k^2}{\sqrt{\beta + k^{10}}}$$

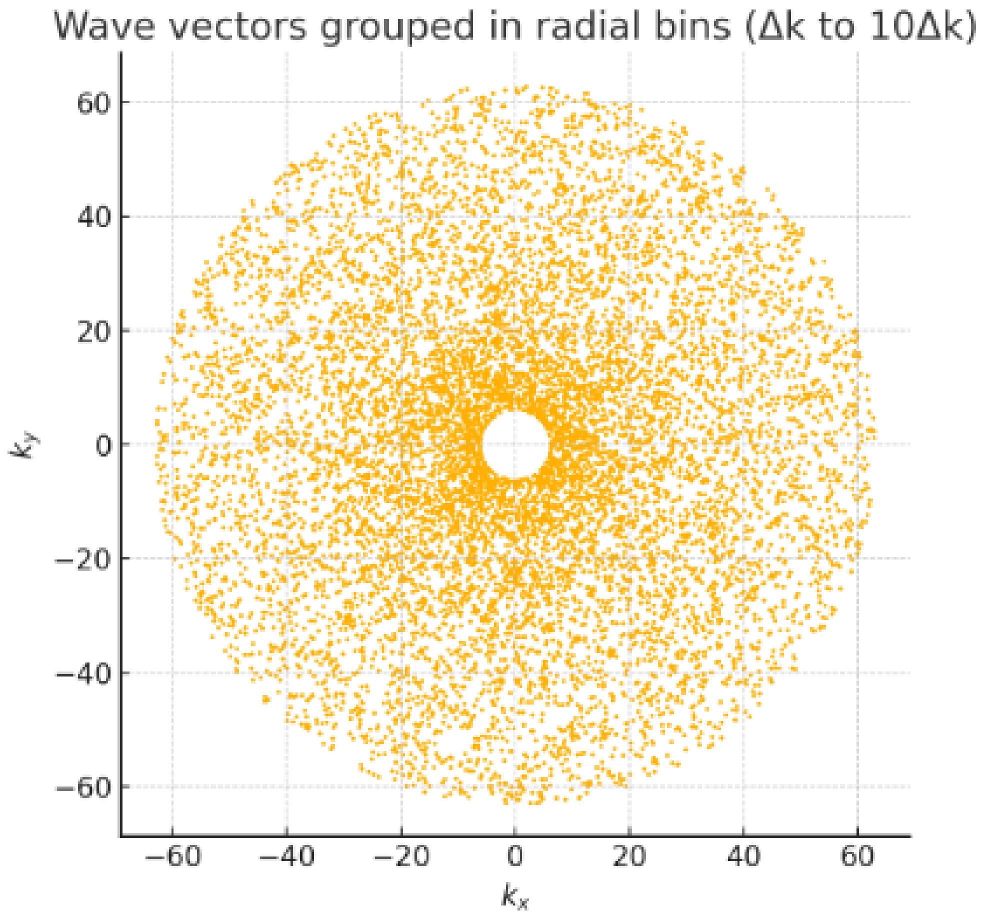


Generating hyperuniform structures
HU scalar fields

$$f(\mathbf{r}) = C \sum_{i=1}^{N} A(k_i) \cos((\mathbf{k}_i + K\hat{\mathbf{k}}_i) \cdot \mathbf{r} + \phi_i) \Delta k_i^d$$







$$\tilde{\psi}_{stealthy}(\bar{k}) = \tilde{\psi}_{hyperuniform}(\bar{k} - K), \text{ for } \breve{k} \geq K$$

Generating hyperuniform structures
HU scalar fields

- Our findings demonstrate that Gaussian random fields offer a simple and effective method for producing disordered hyperuniform or stealthy scalar fields over large length scales. This approach can serve as a practical guide for experimentalists aiming to create new hyperuniform materials using techniques like stereolithography or 3D printing.

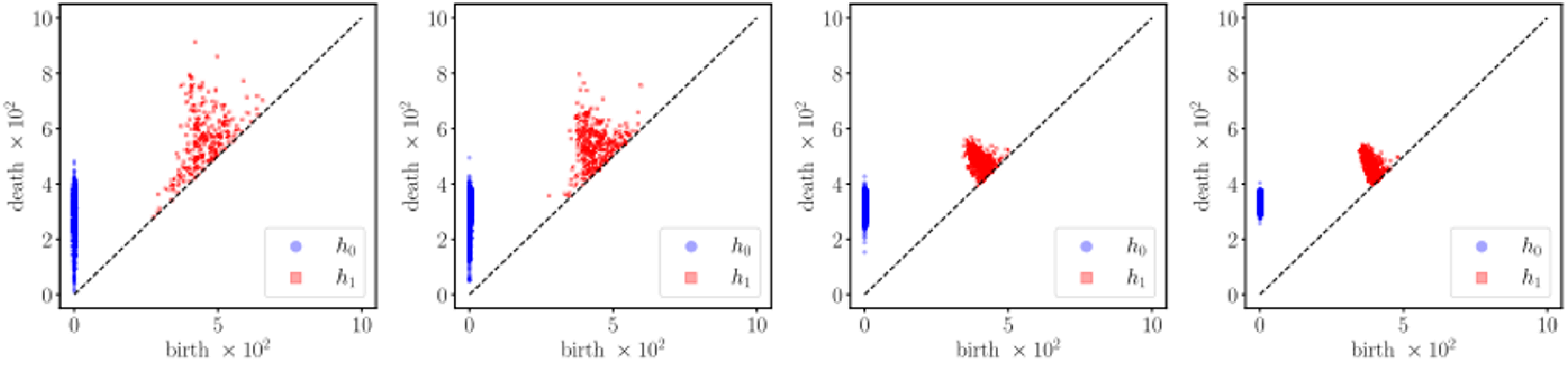
Persistent Homology Analysis

Persistent Homology Analysis

Persistent Homology and Topological Analysis

- A Vietoris–Rips complex is built by growing balls of radius *r* around each point
- Homological features are computed: *h0* (connected components), *h1* (loops)
- Persistence diagrams and histograms of death times are analyzed to capture topological features.

<u>Task:</u> Use *ripser* or *GUDHI* to compute persistent homology and analyze the resulting diagrams



Persistent Homology Analysis
Topological Comparison and Robustness to Finite Sampling

- Wasserstein distances *W1*, *W2* are used to compare persistence diagrams:
  1. Across different pattern parameters
  2. Between full systems and cropped subsets

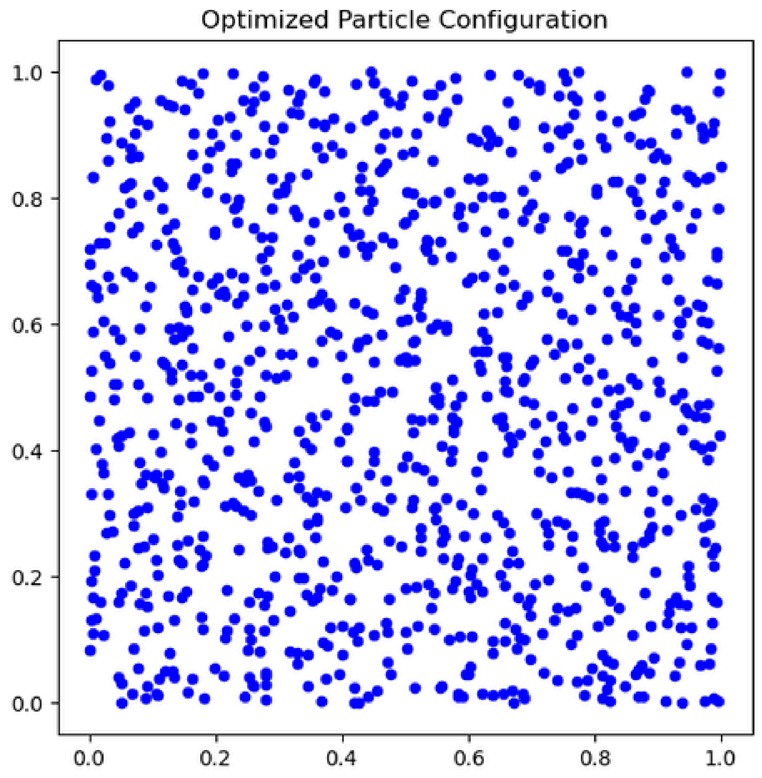
$$W_p(A,B) = \inf_{\gamma} \left(\sum_{\mathbf{x} \in A} (||\mathbf{x} - \gamma(\mathbf{x})||_q)^p \right)^{1/p},$$

- The study shows that HU properties are preserved in subsets above a certain size

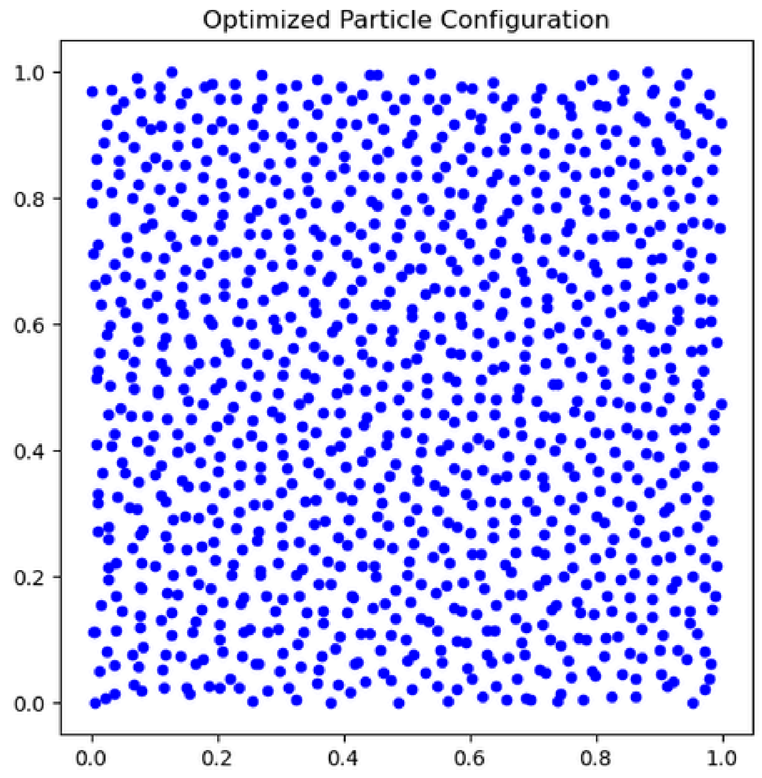
Task: Compare diagrams using Wasserstein distance, identify HU regions in parameter space, and analyze robustness to partial data

Persistent Homology Analysis
Parameters

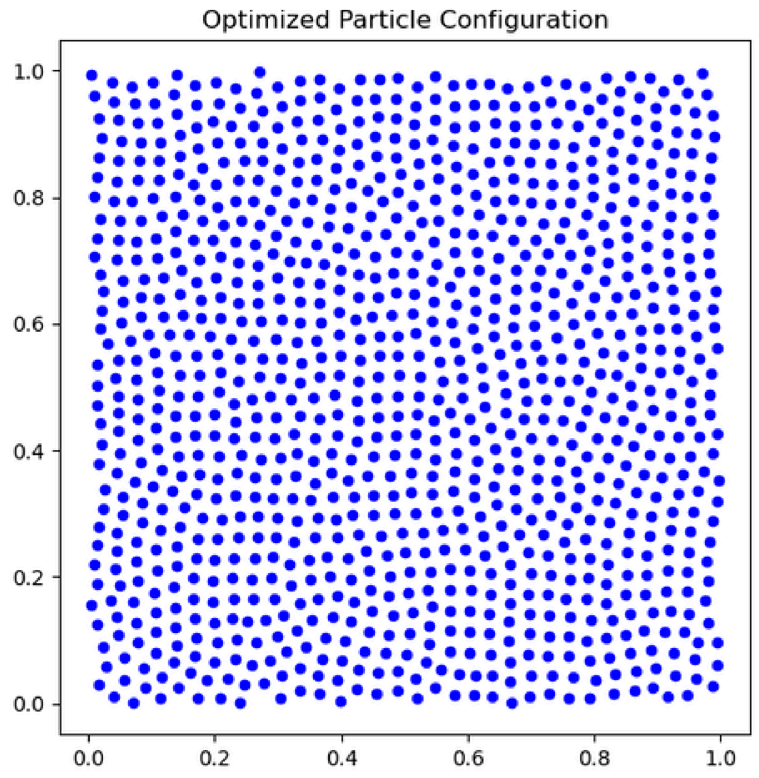
1. Generates and optimizes points configurations with varying parameters (α, k) to match a target structure factor using BFGS optimization.
2. Saves each result for analysis and comparison.



$\alpha=1$
$k=20\pi$



$\alpha=10$
$k=40\pi$

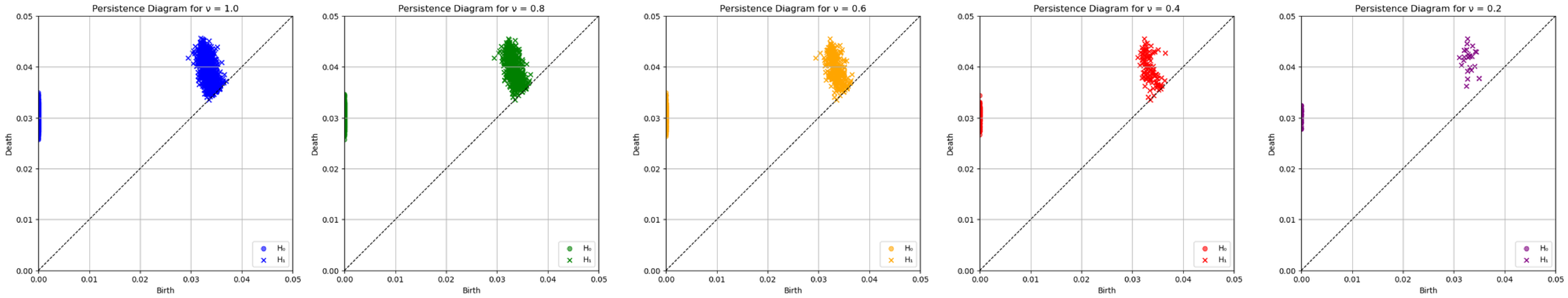


$\alpha=100$
$k=60\pi$

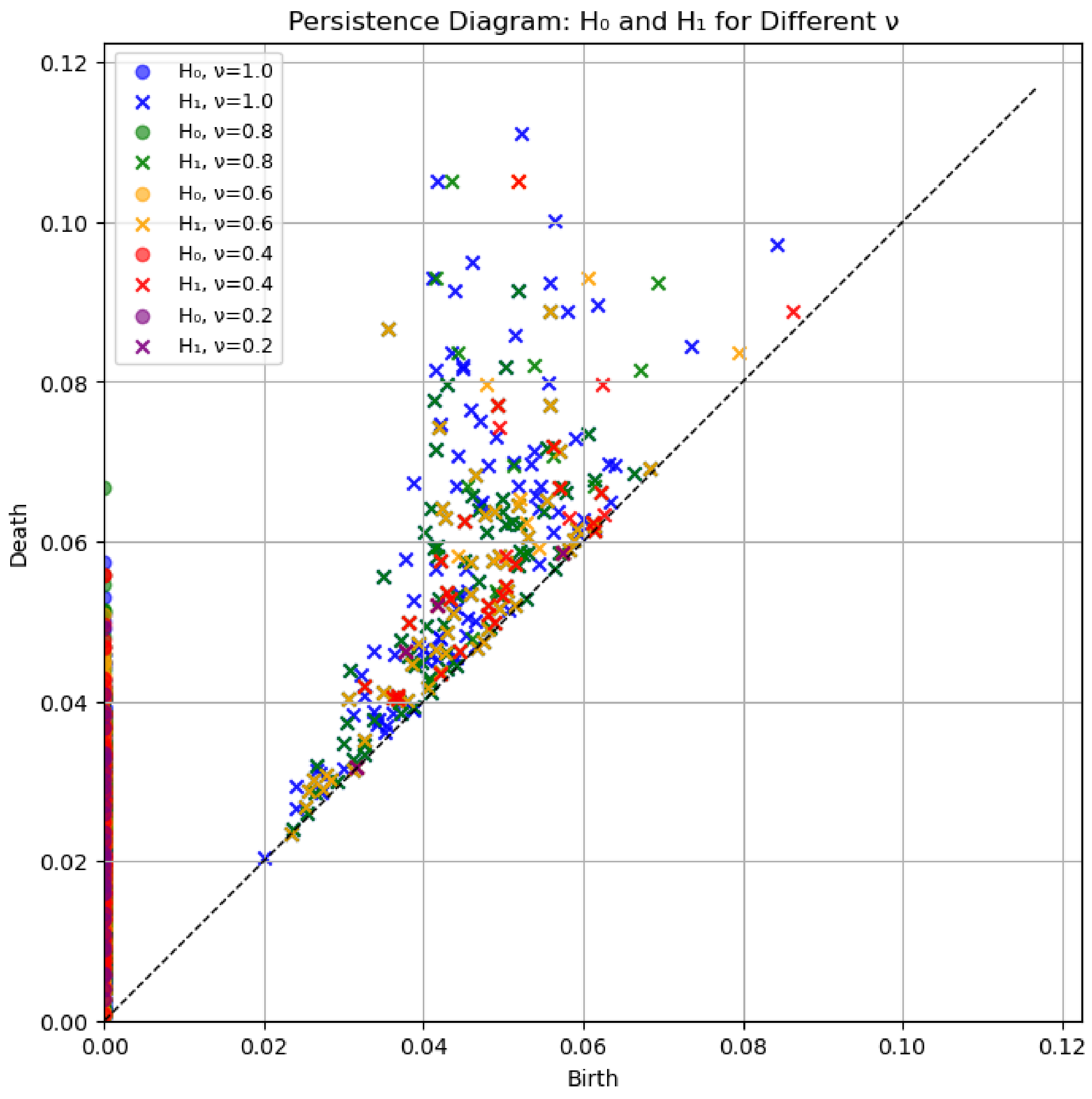
Persistent Homology Analysis

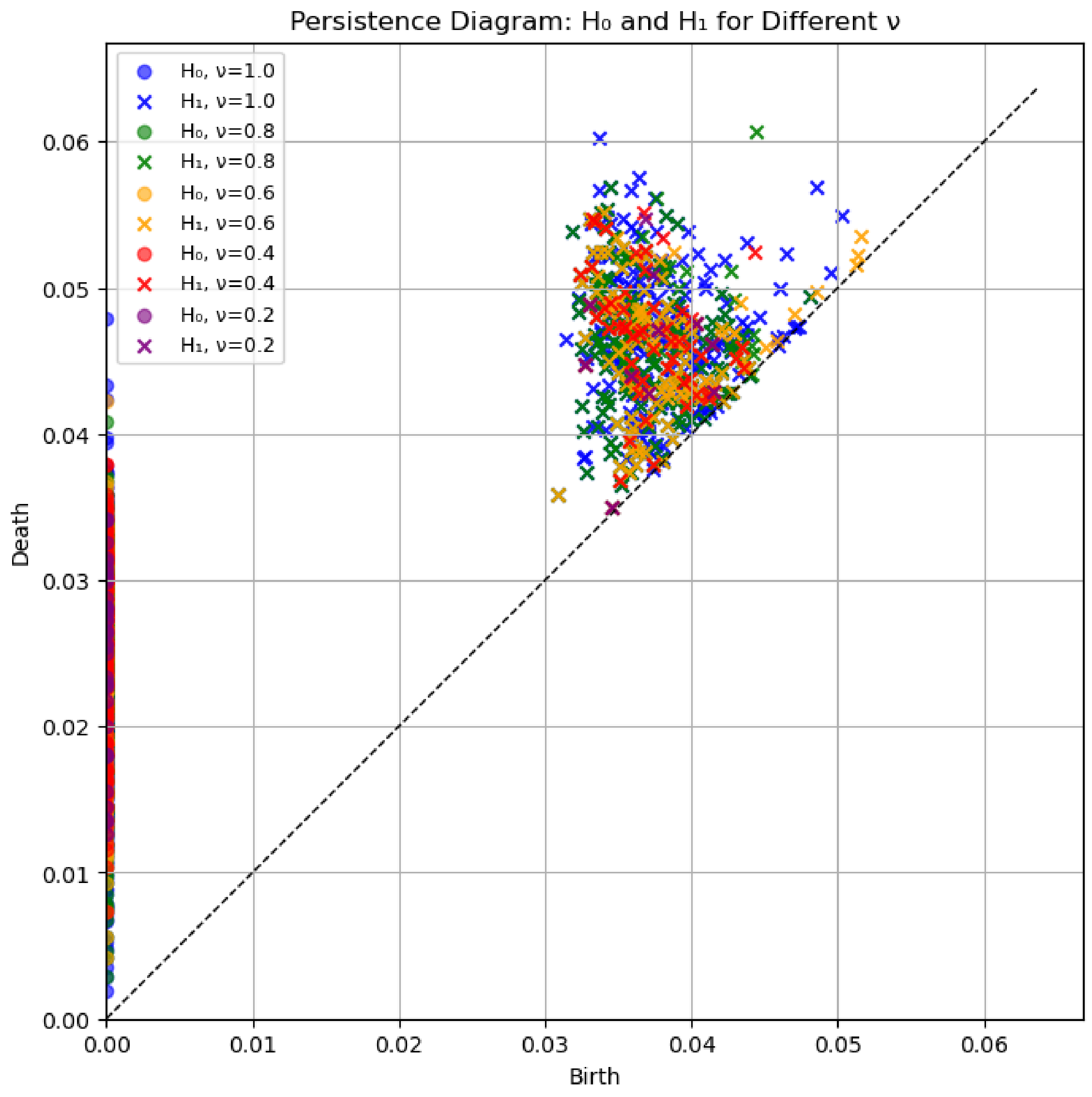
Persistence

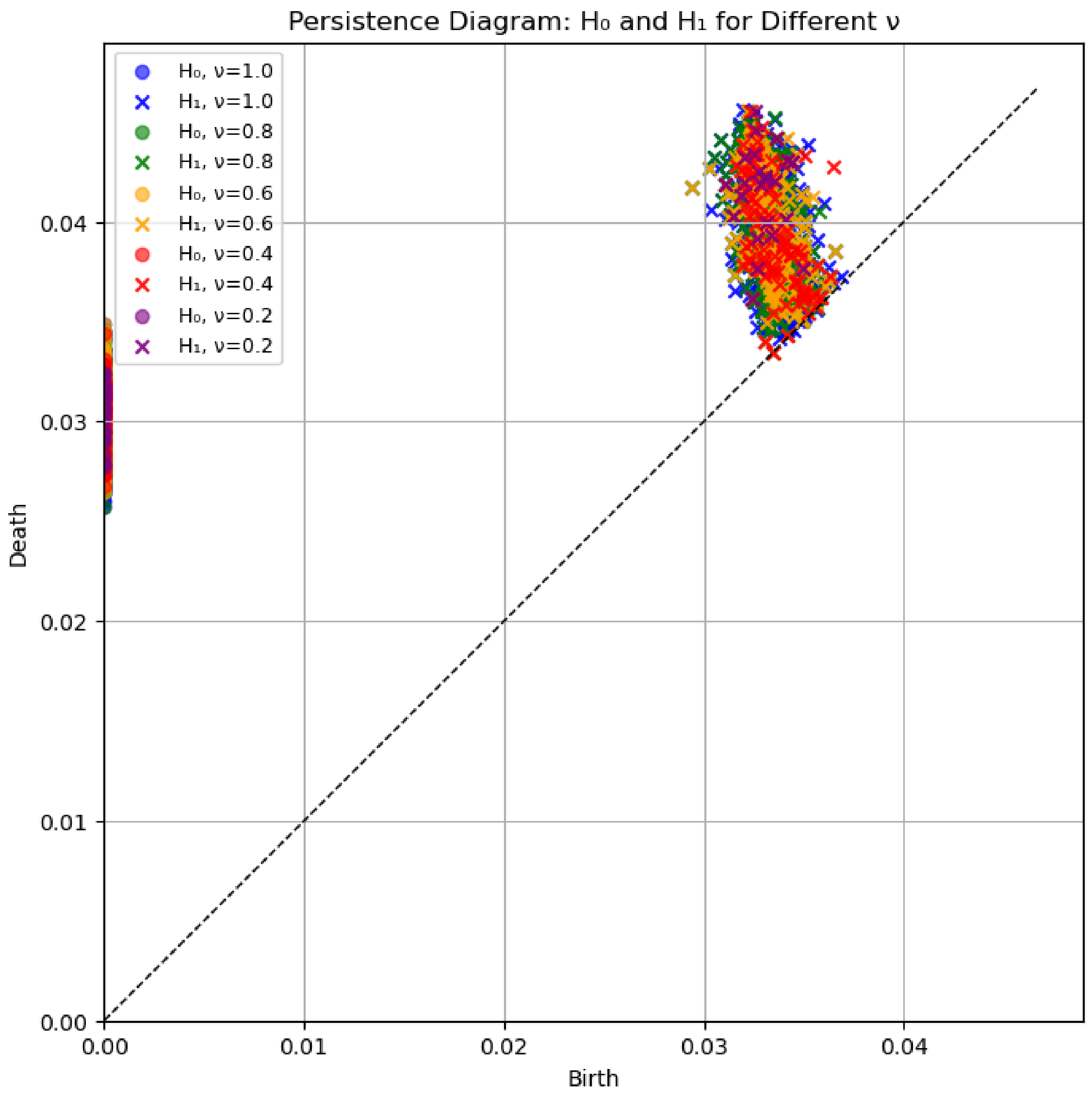
The result is a set of persistence diagrams for different values of ν. Each diagram shows when connected components (H_0) and loops (H_1) appear (“birth”) and disappear (“death”) in the point cloud patches. This reveals how the topological features change with scale, helping analyze the data’s shape and structure.



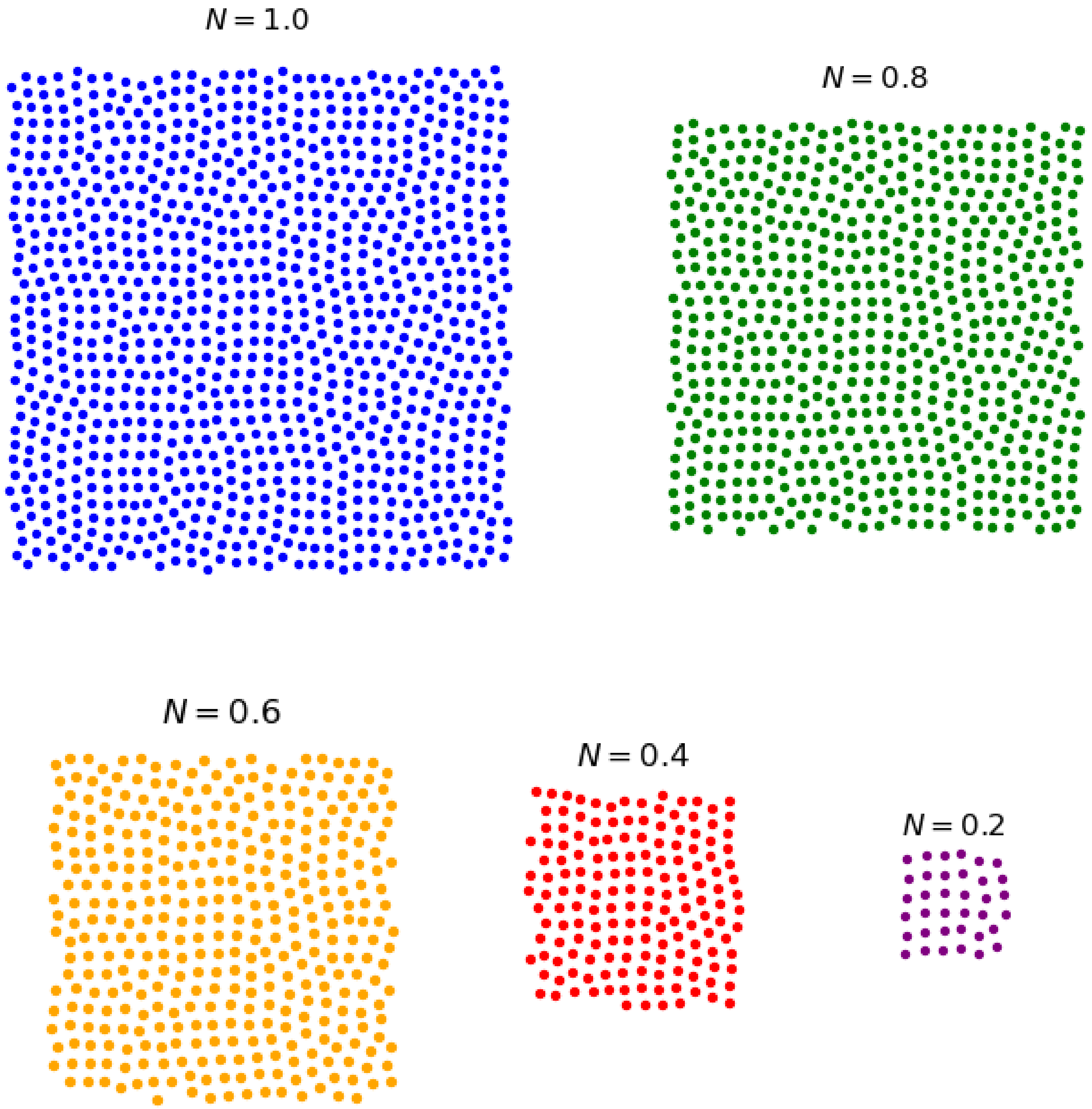
Persistent Homology Analysis
Persistence



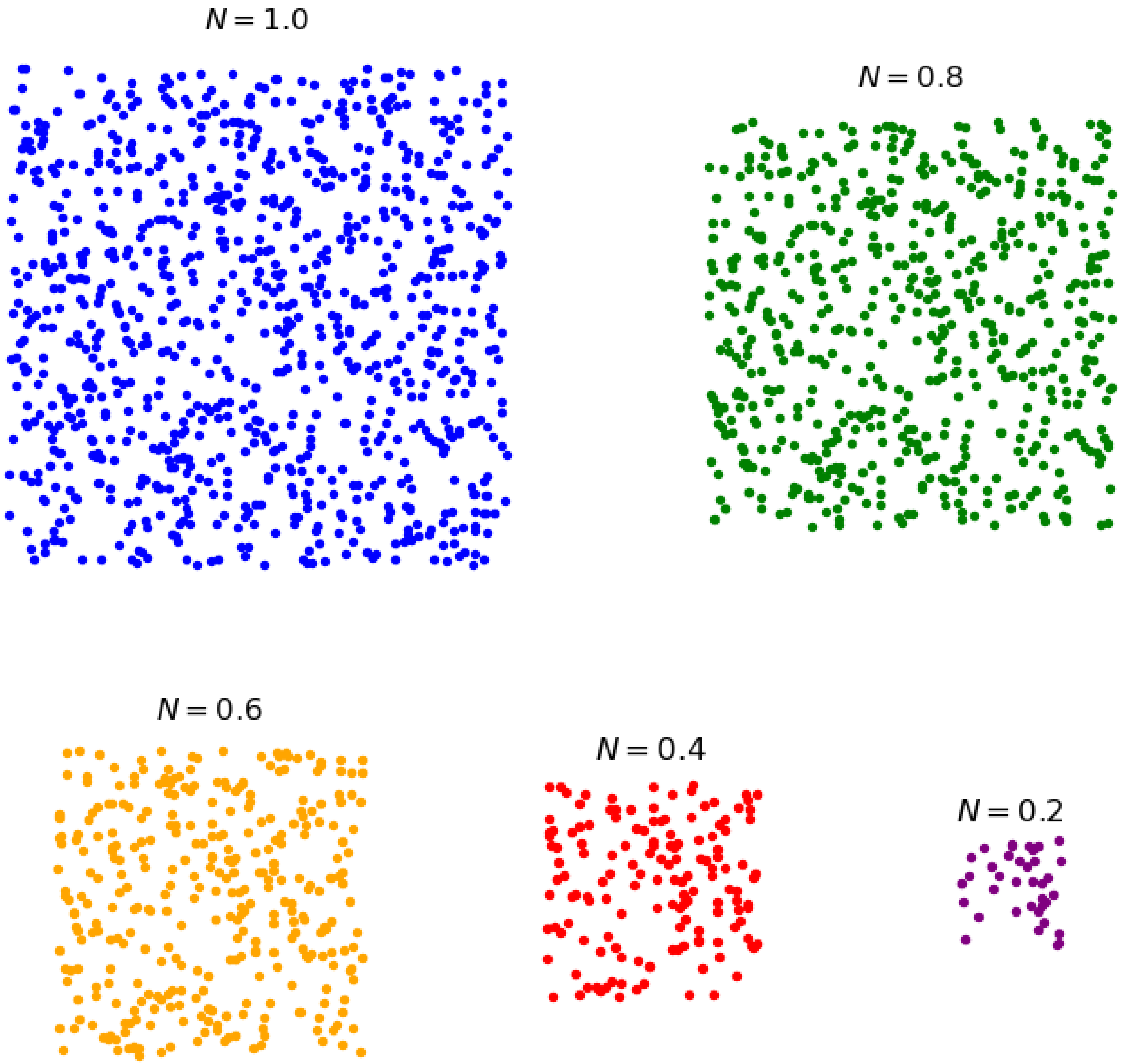




α=100 K=60pi

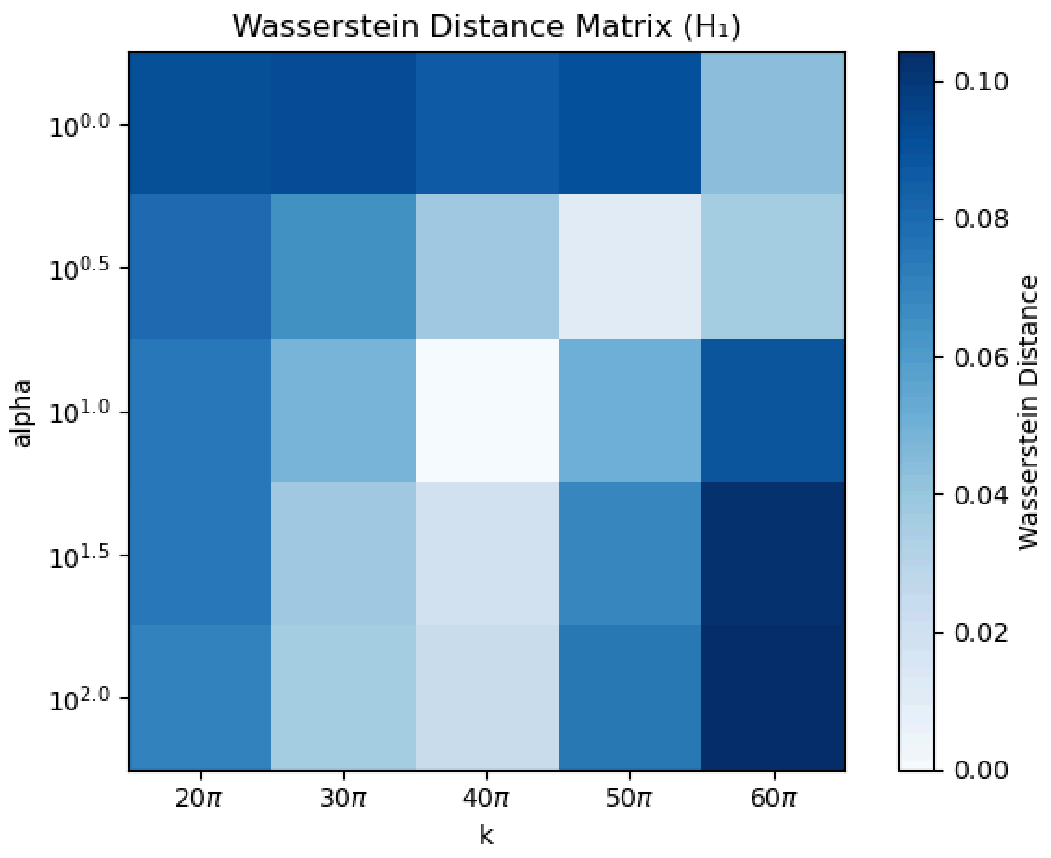


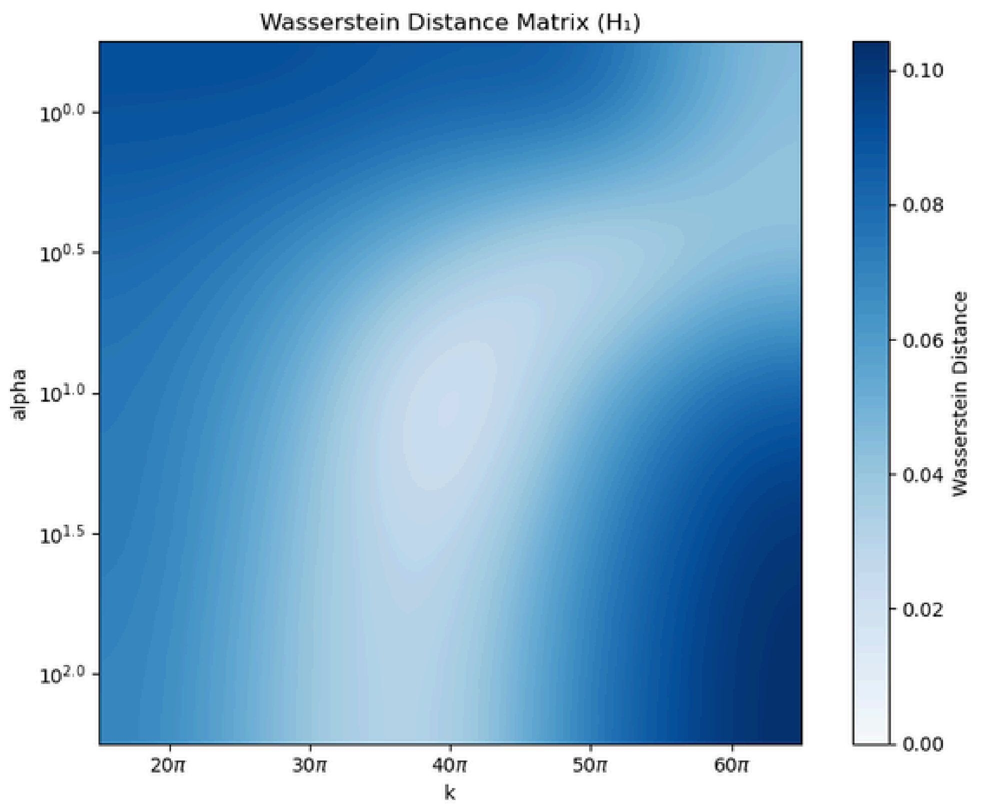
α=1 K=20pi

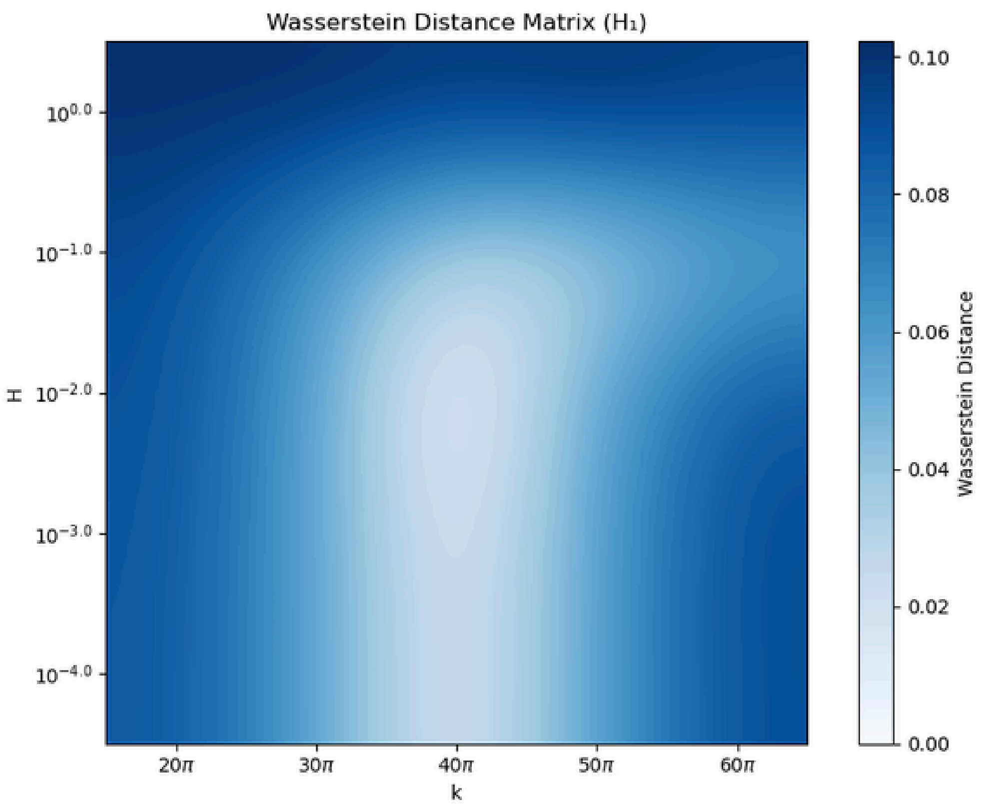


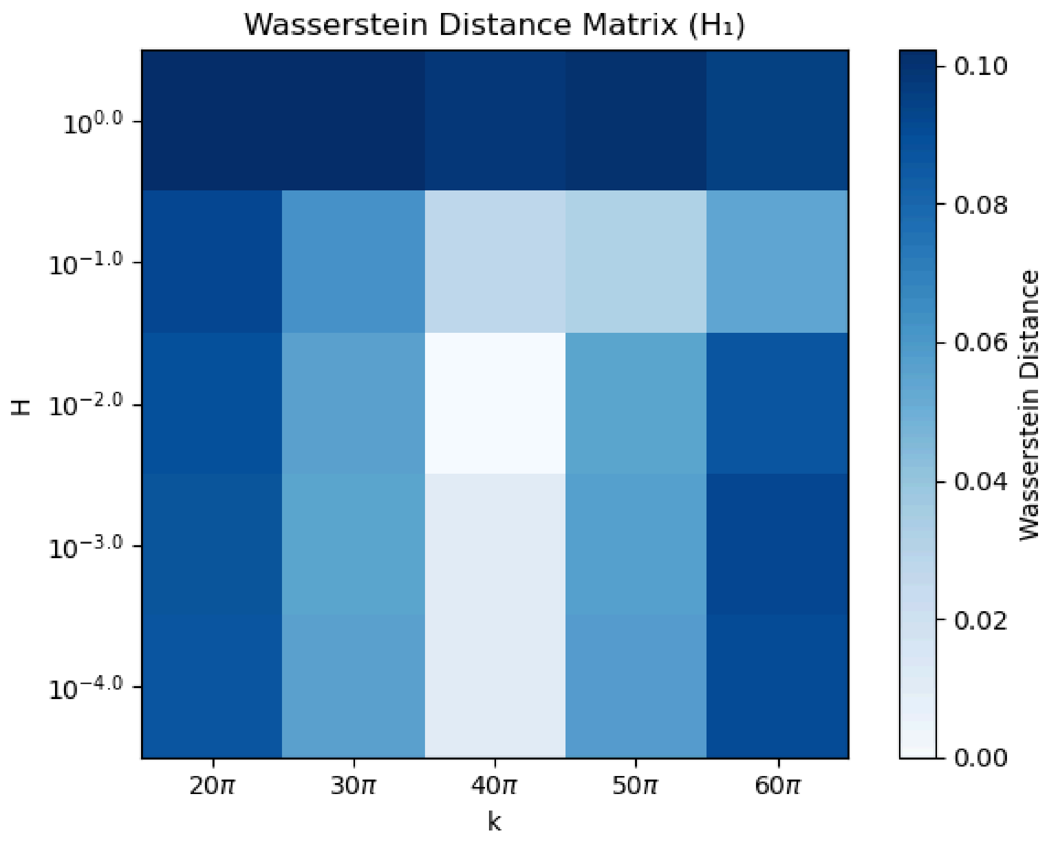
Persistent Homology Analysis
Wasserstein distance

The graphics show how similar the topological features of different point configurations are to a reference configuration Wasserstein distances indicate more similar topological structures. This allows assessing how parameter changes affect the topology of the data.





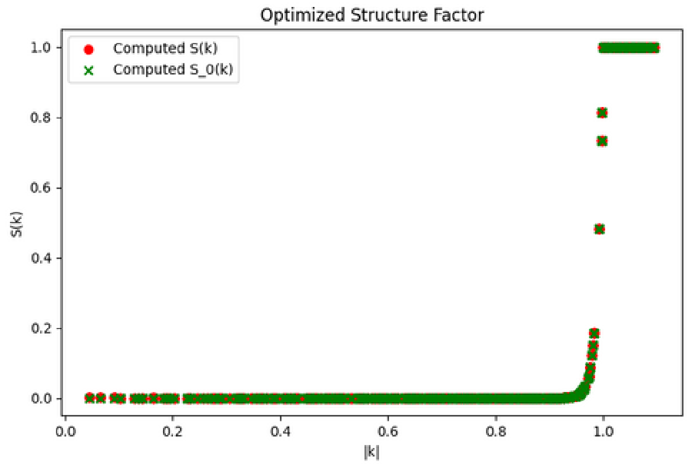


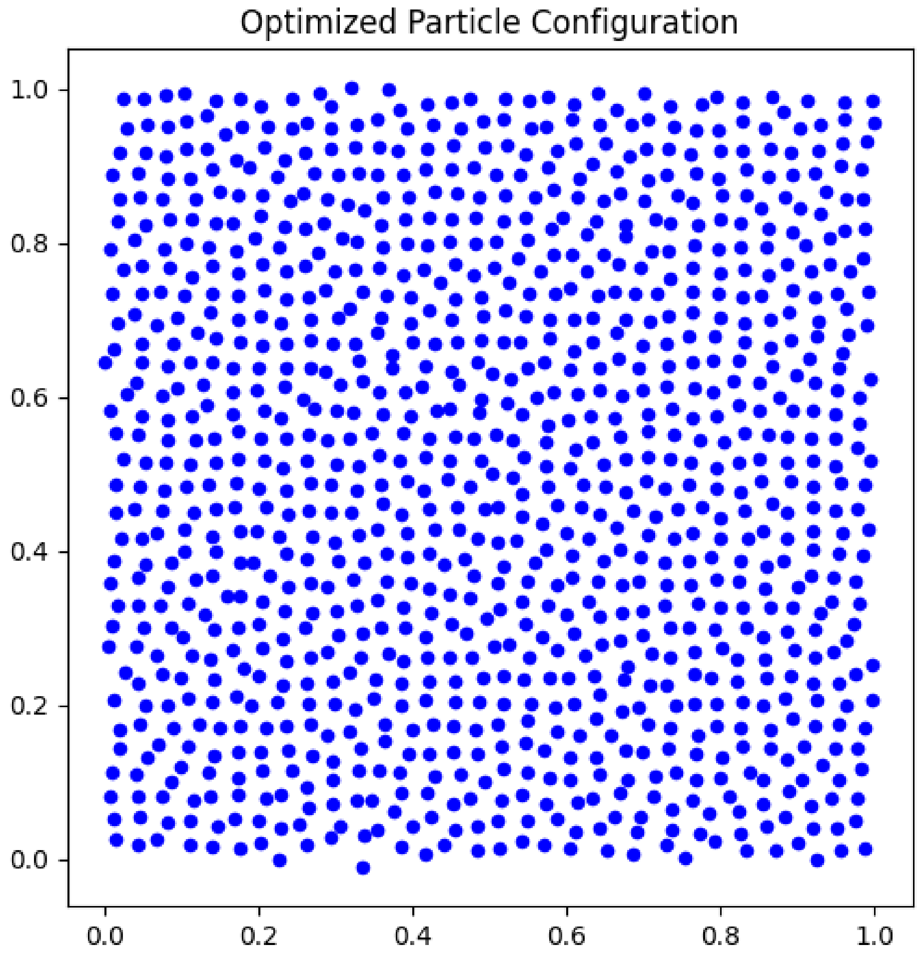


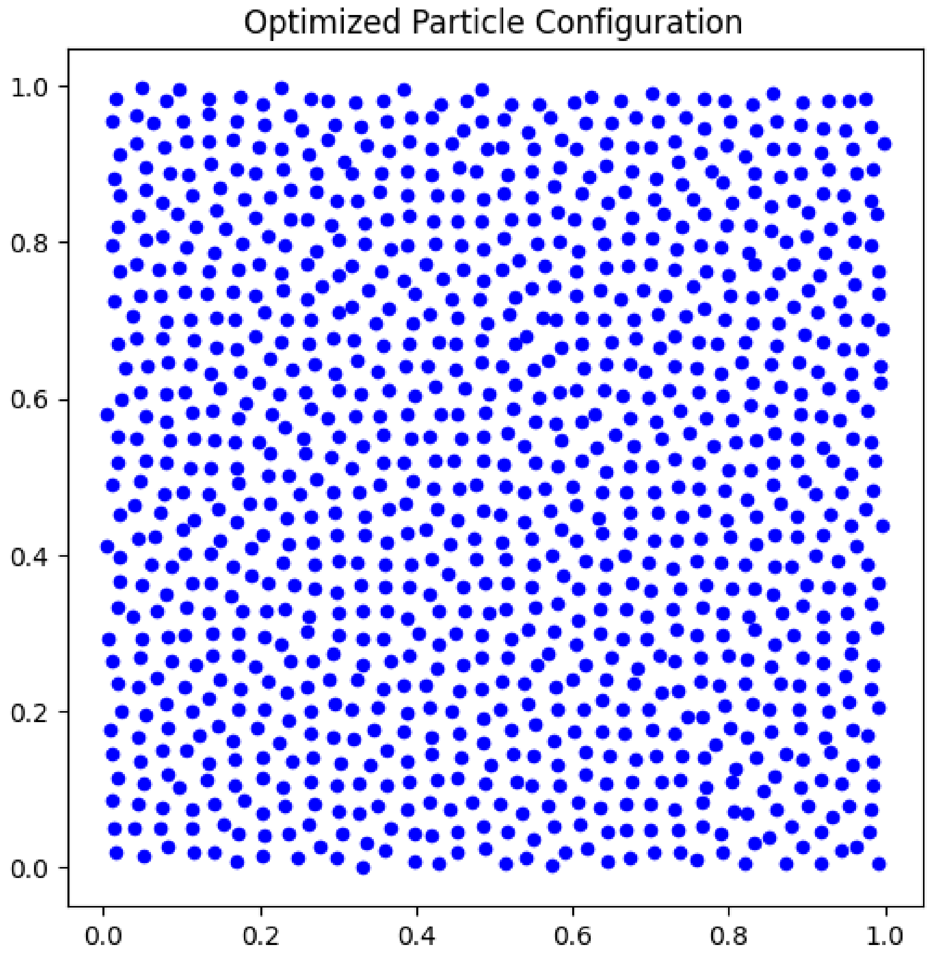
Persistent Homology Analysis

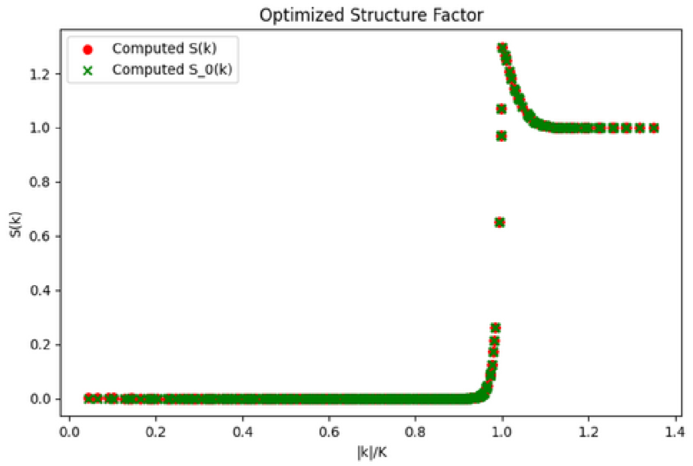
Analysis of modified a objective structure factor

- We may slightly change the objective structure factor with a bump (as indeed is often observed in real-world experiments [Tor18])
- We have already seen two point configurations with the only difference being a bump - visually no discernable difference









Persistent Homology Analysis
Analysis of modified a objective structure factor

- Can we quantify the difference using the persistent homology?
- We compute for random initial perturbations the mean and variance within the no bump group to eachother and between the bump and no bump group

	no bump vs. no bump	no bump vs. bump
h_0	$[0.192; 1.1 \cdot 10^{-3}]$	$[0.264; 3.3 \cdot 10^{-3}]$
h_1	$[0.231; 6.0 \cdot 10^{-4}]$	$[0.258; 6.3 \cdot 10^{-4}]$

- Conclusion: The difference is small but discernible, especially in the h0 component.
- Illusiveness of hyperuniformity can effectively be analyzed using the persistent homology!

Summary

Summary

- Hyperuniform systems suppress large-scale density fluctuations, shown by a vanishing structure factor at small wave vectors.
- We created various parameterized hyperuniform point patterns and scalar fields.
- We benchmarked different generation techniques.
- We applied persistent homology to study diverse hyperuniform structures
- Our generation methods are versatile, allowing us to produce other structures with appropriate parameter choices, potentially connecting to quasiperiodic and Gaussian random field models.

Literature

Literature

- [Tor18] Salvatore Torquato. Hyperuniform states of matter. Physics Reports, 745:1–95, 2018. Hyperuniform States of Matter.
- [SSDV24] Marco Salvalaglio, Dominic J. Skinner, Jörn Dunkel, and Axel Voigt. Persistent homology and topological statistics of hyperuniform point clouds. Phys. Rev. Res., 6:023107, May 2024.
- [MT17] Zheng Ma and Salvatore Torquato. Random scalar fields and hyperuniformity. Journal of Applied Physics, 121(24):244904, 2017.
- [SCM24] Aaron Shih, Mathias Casiulis, and Stefano Martiniani. Fast generation of spectrally shaped disorder. Phys. Rev. E, 110:034122, Sep 2024.
- [FIN24] FINUFFT Contributors. FINUFFT Mathematical Formulation. Accessed: 2025-06-04. Flatiron Institute Nonuniform Fast Fourier Transform (FINUFFT). 2024. url: https://finufft.readthedocs.io/en/latest/math.html.